



Acoustics of porous media

Lecture 4: Transfer Matrix Method

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Presentation

A simple method to determine transmission of panels where the properties of each layer are known

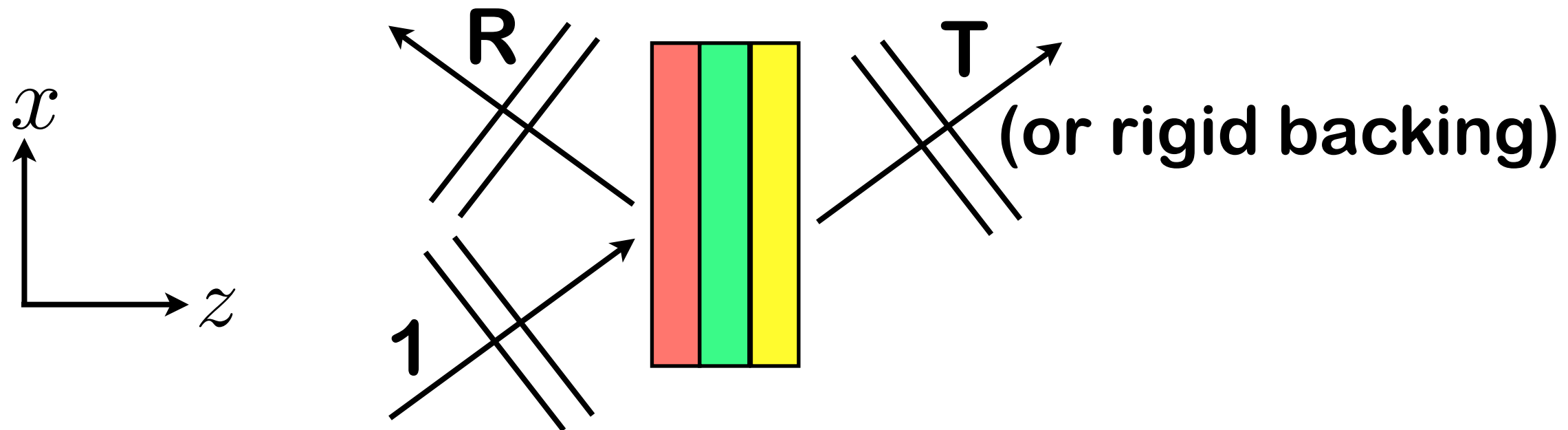
Main assumptions

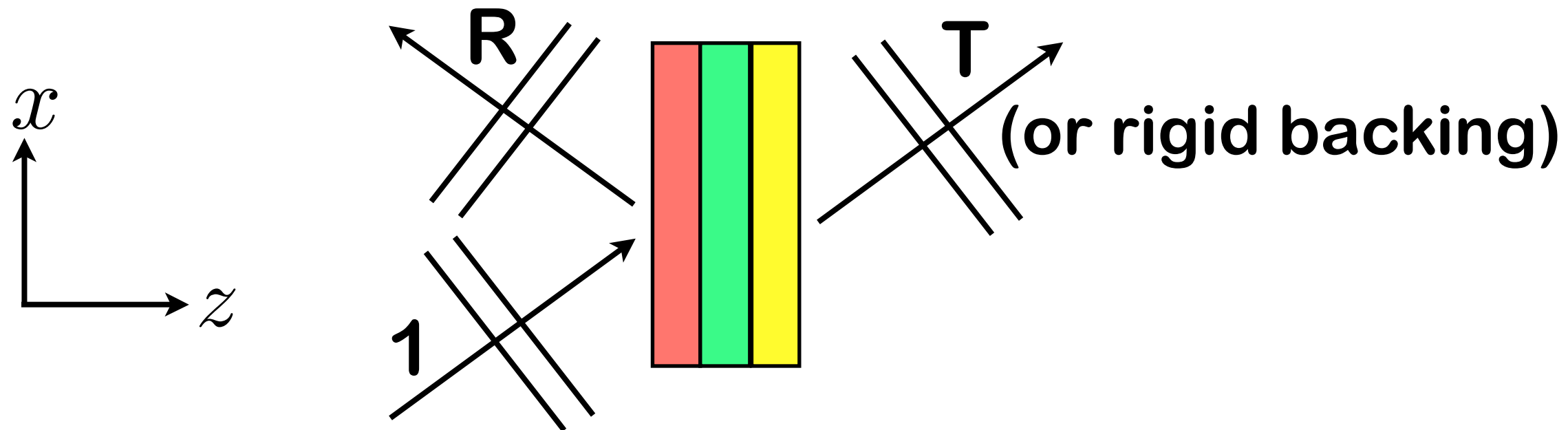
- Multilayered structures
 - Infinite lateral dimensions
 - Plane interfaces
- Layers invariant by rotation around thickness direction
- Excitation by an airborne plane wave

Applications

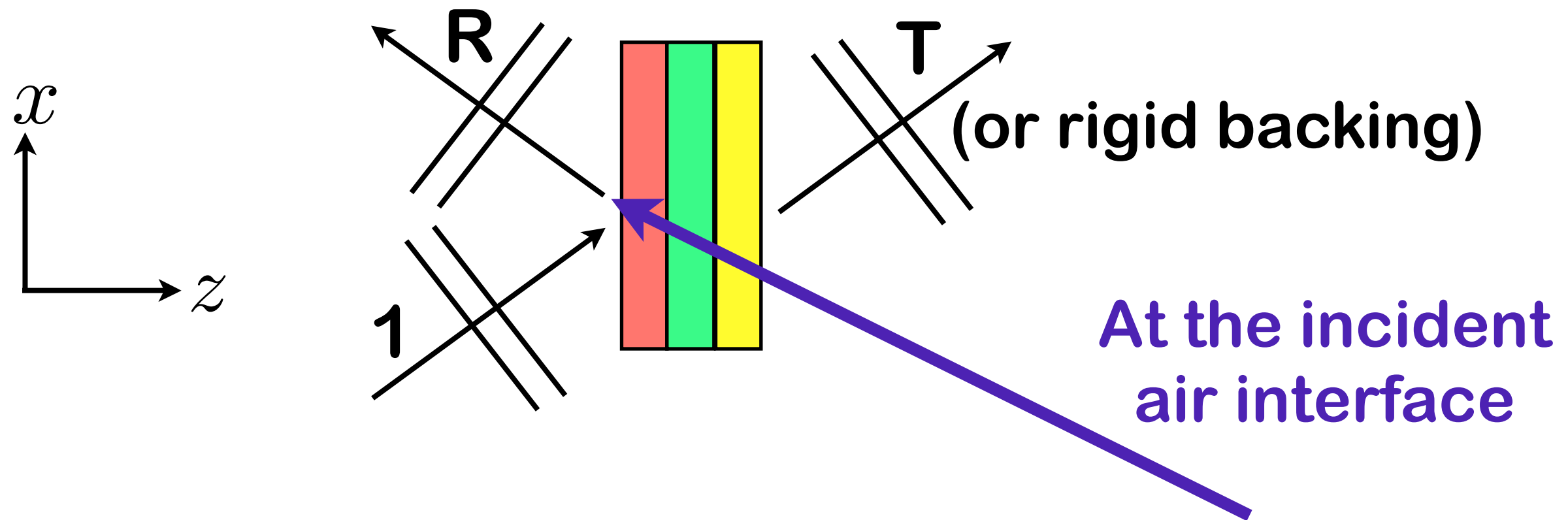
- Building Acoustics
- Automotive applications ...

- Different type of modelling
 - Equivalent fluid model / Limp model
 - Only one compressional wave
 - Dissipative «classical» fluid
 - Elastic solid model (Biot)
 - Two compressional waves
 - One shear wave
- Coupling conditions
 - Elastic solid
 - Fluid
 - Interface between two porous media





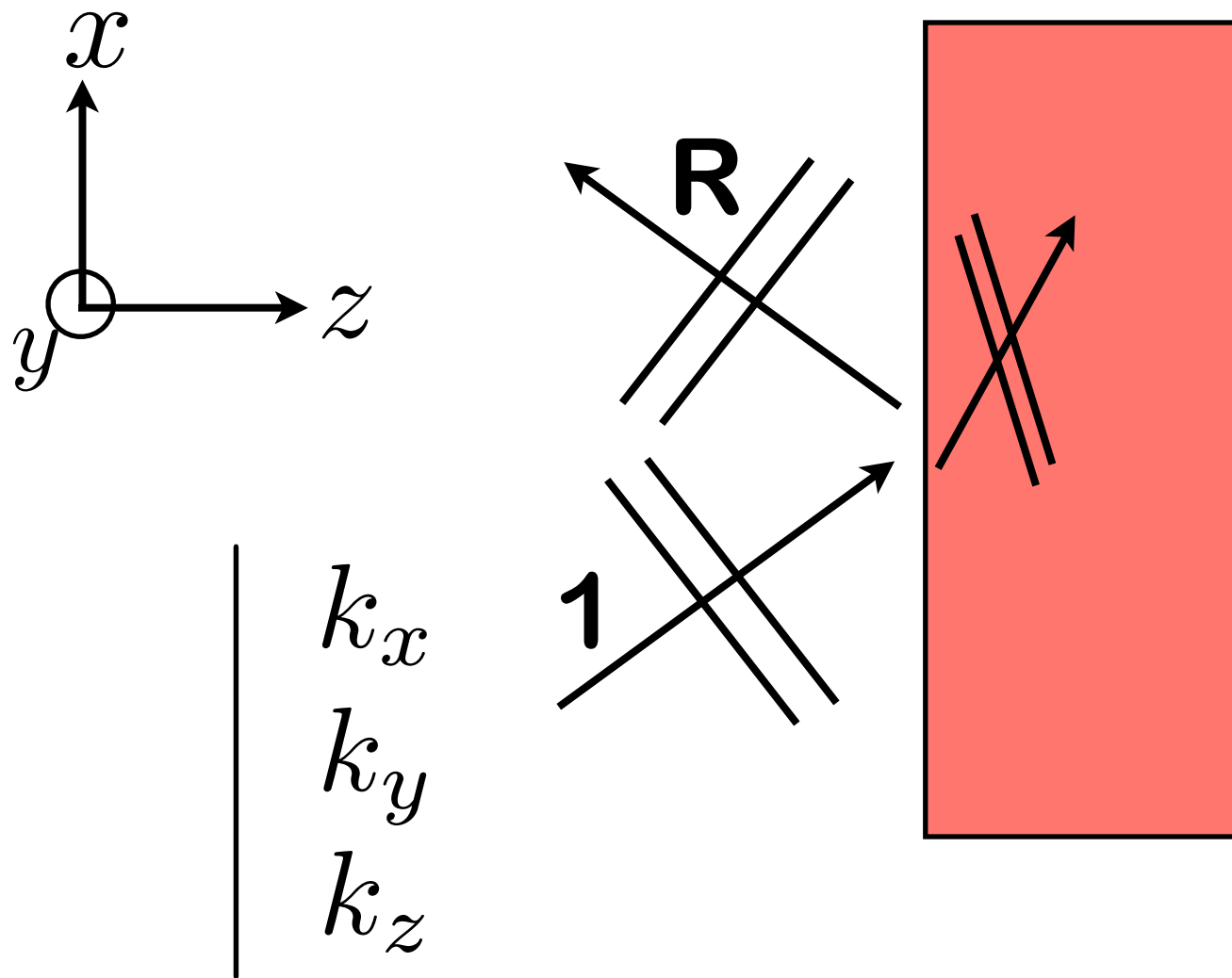
$$R = \frac{Z_s - Z_{air}}{Z_s + Z_{air}}, \quad Z_s = \frac{p_s}{v_s}$$

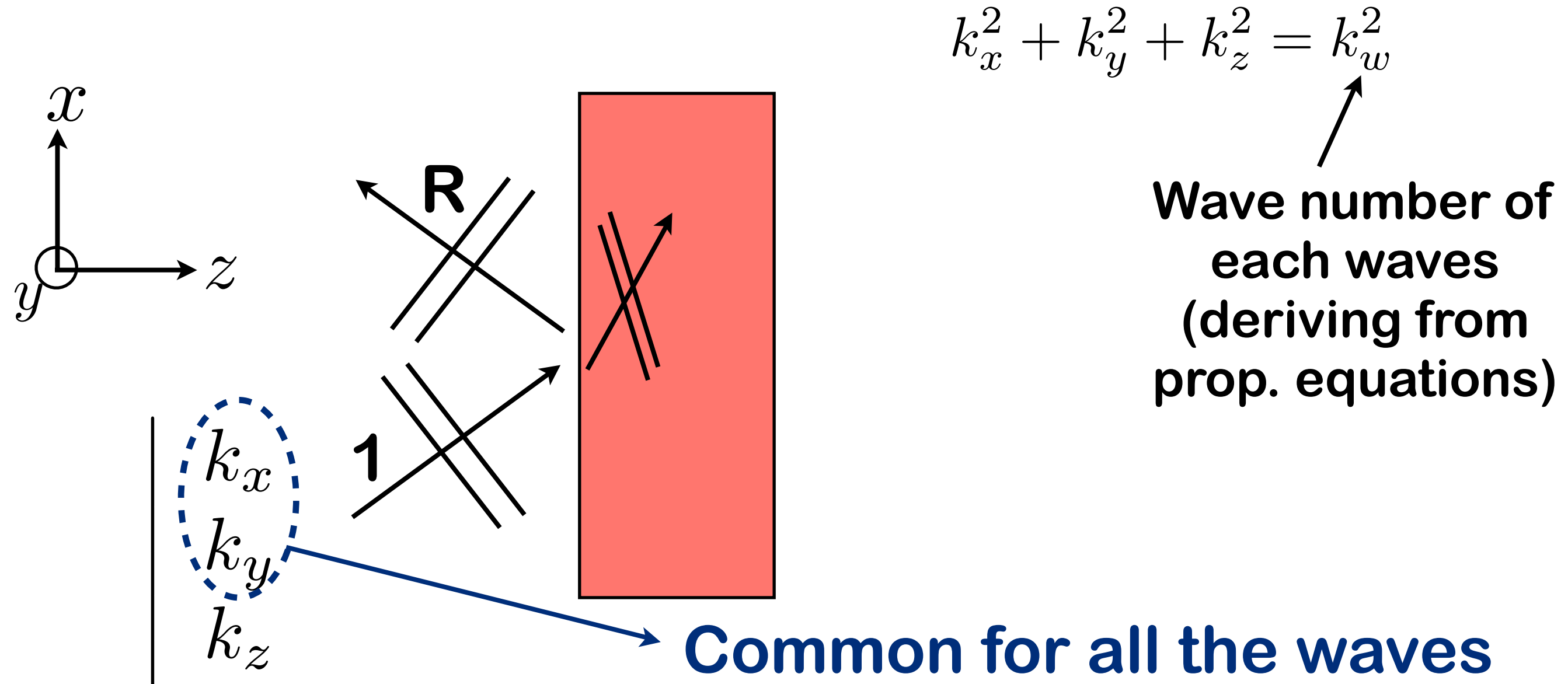


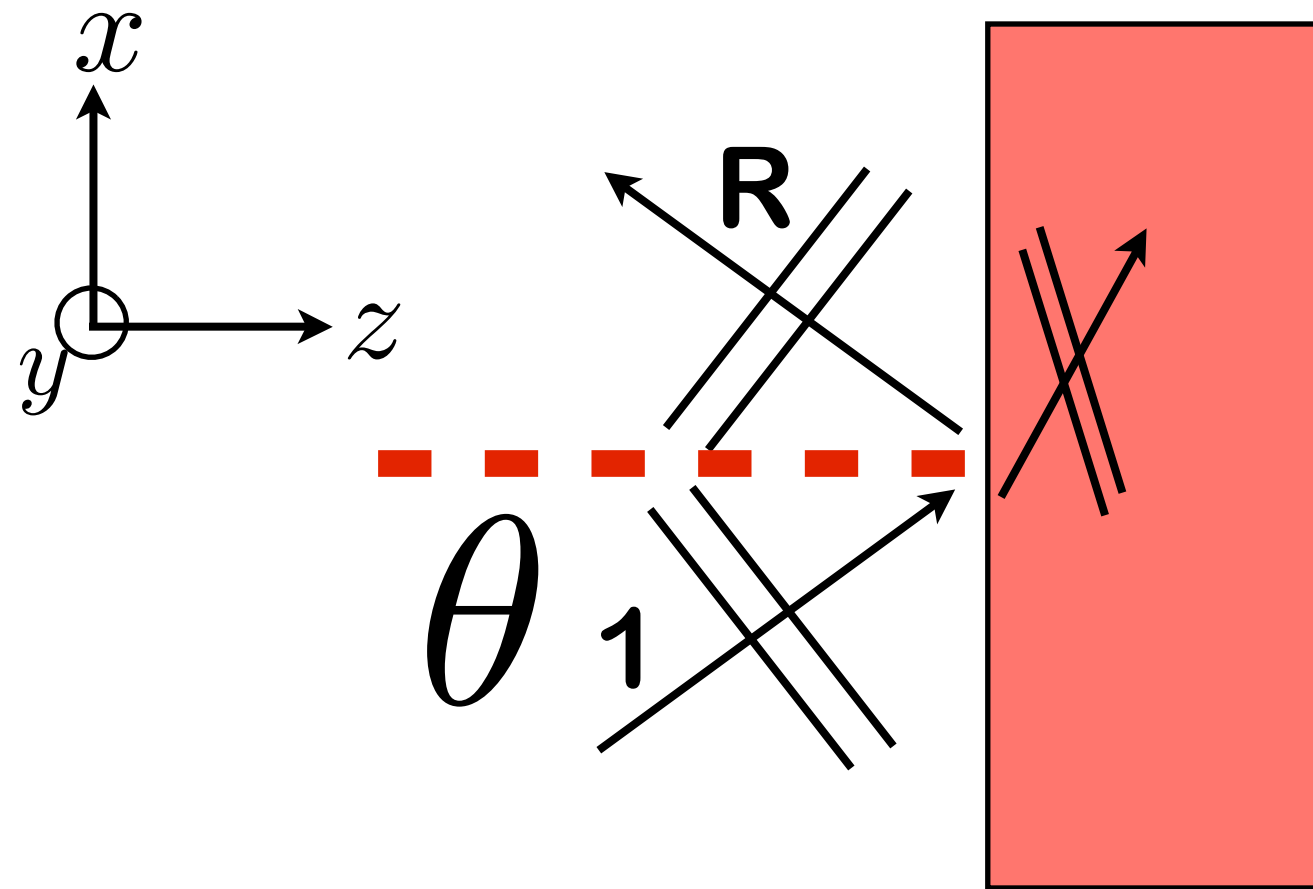
$$R = \frac{Z_s - Z_{air}}{Z_s + Z_{air}}, \quad Z_s = \frac{p_s}{v_s}$$

TMM: A general method to find surface impedance of a multicomponent panel

Few words on Snell-Descartes laws





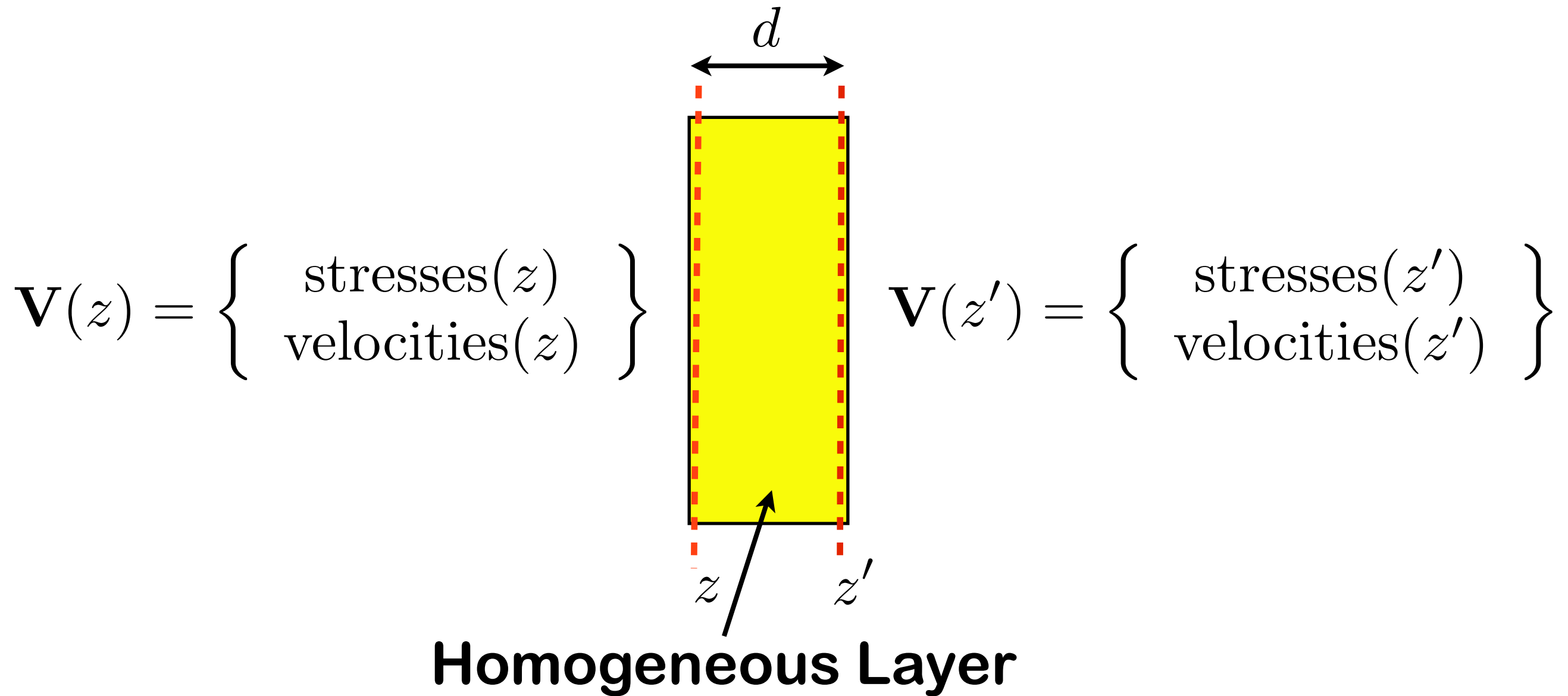


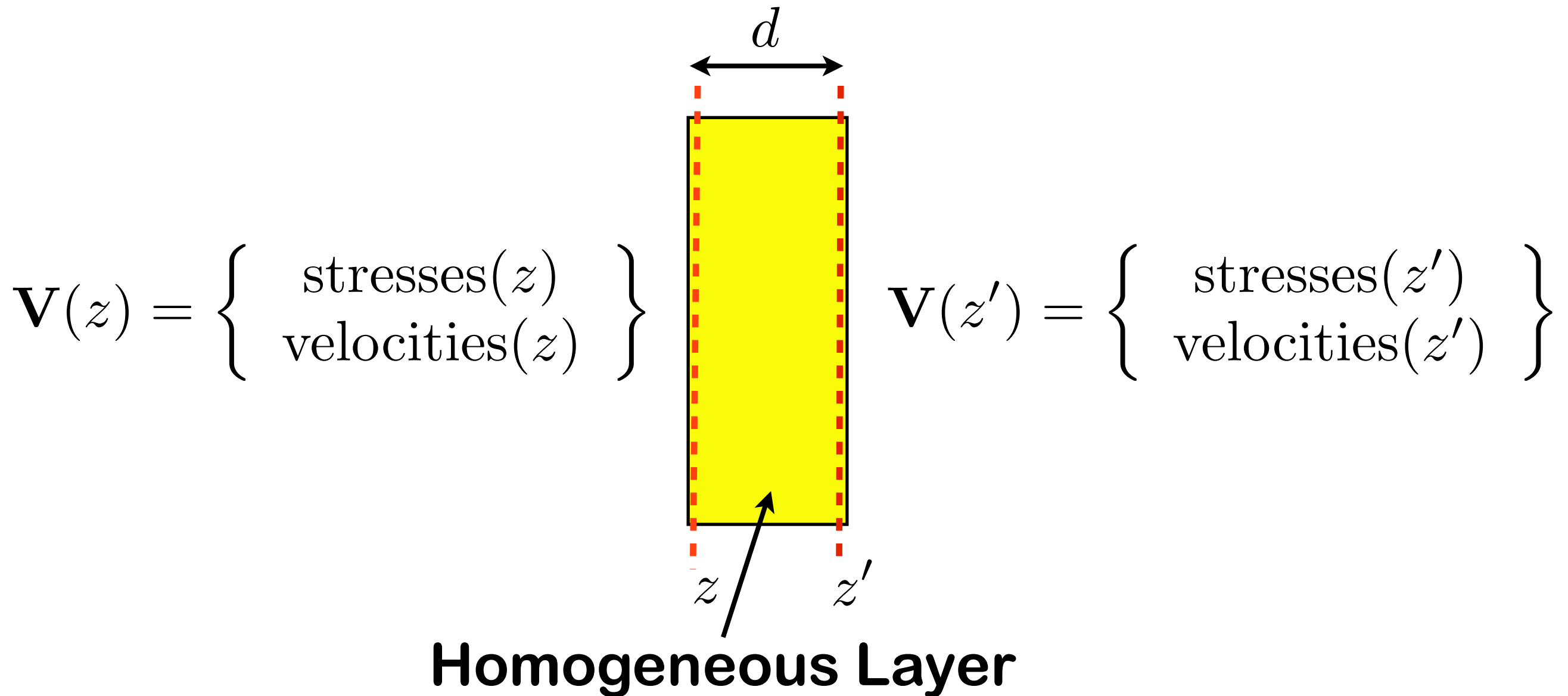
$$k_x^2 + k_y^2 + k_z^2 = k_w^2$$

Wave number of
each waves
(deriving from
prop. equations)

$k_y = 0$ if rotation invariance of the panel

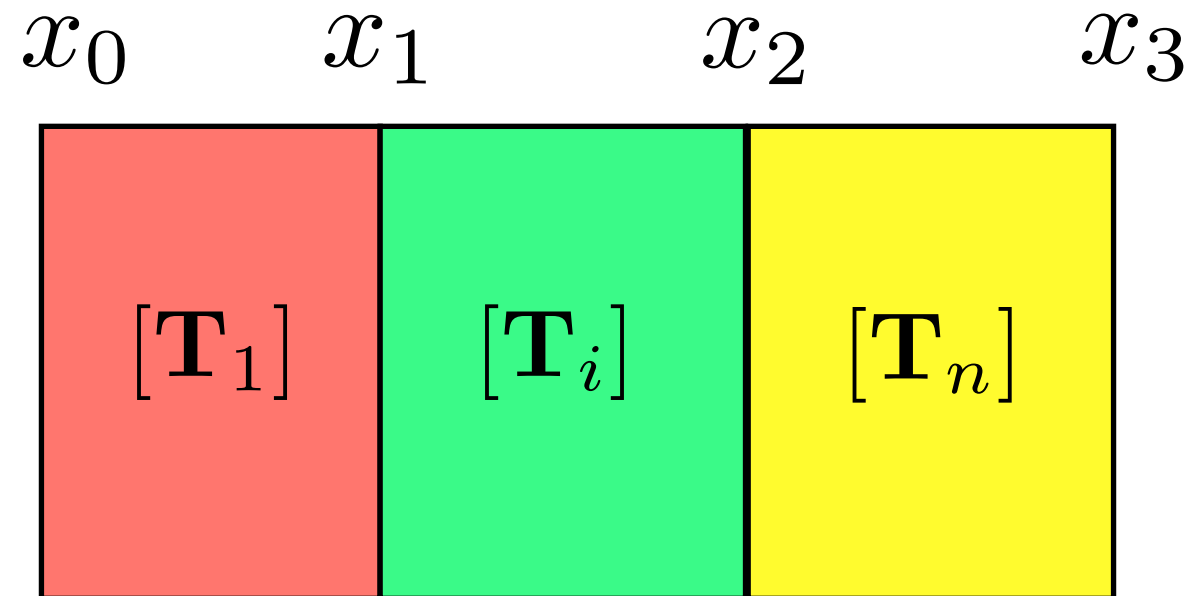
$$\left| \begin{array}{lcl} k_x & = & k_0 \sin(\theta) \\ k_y & = & 0 \\ k_z & = & k_0 \cos(\theta) \end{array} \right.$$



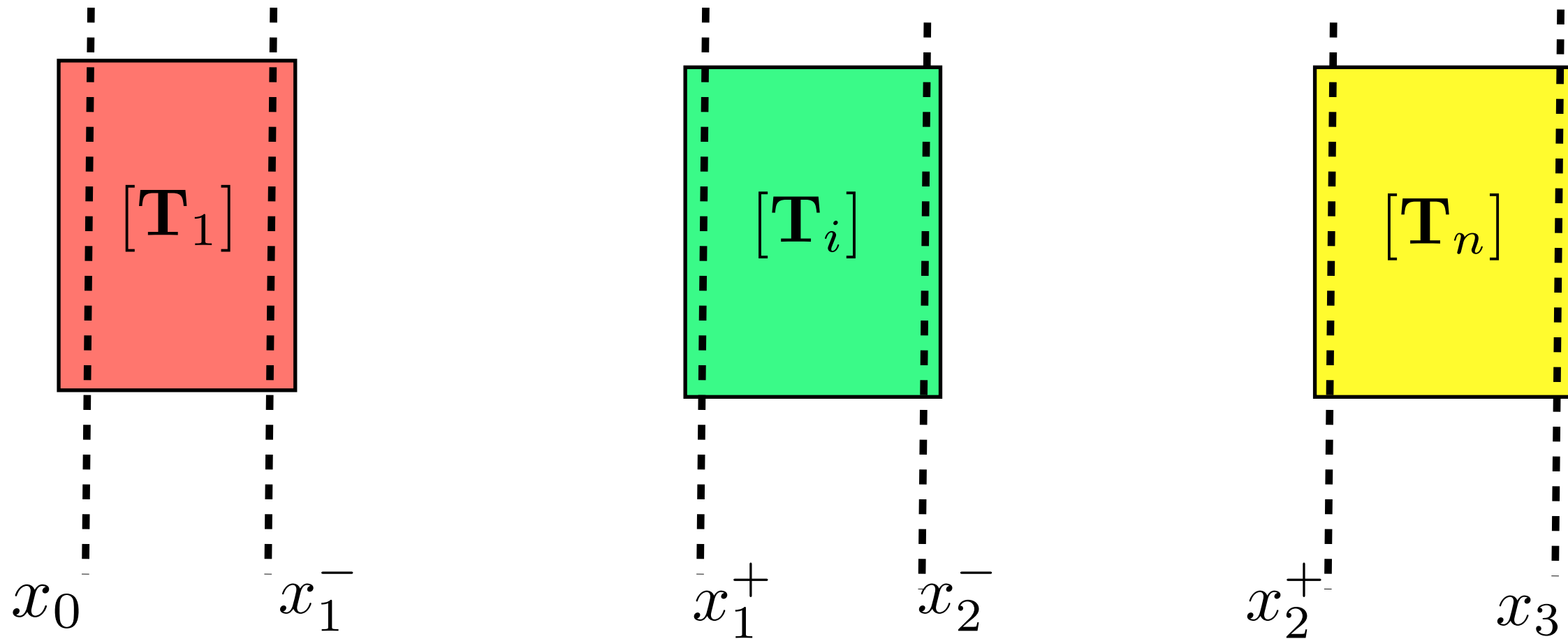


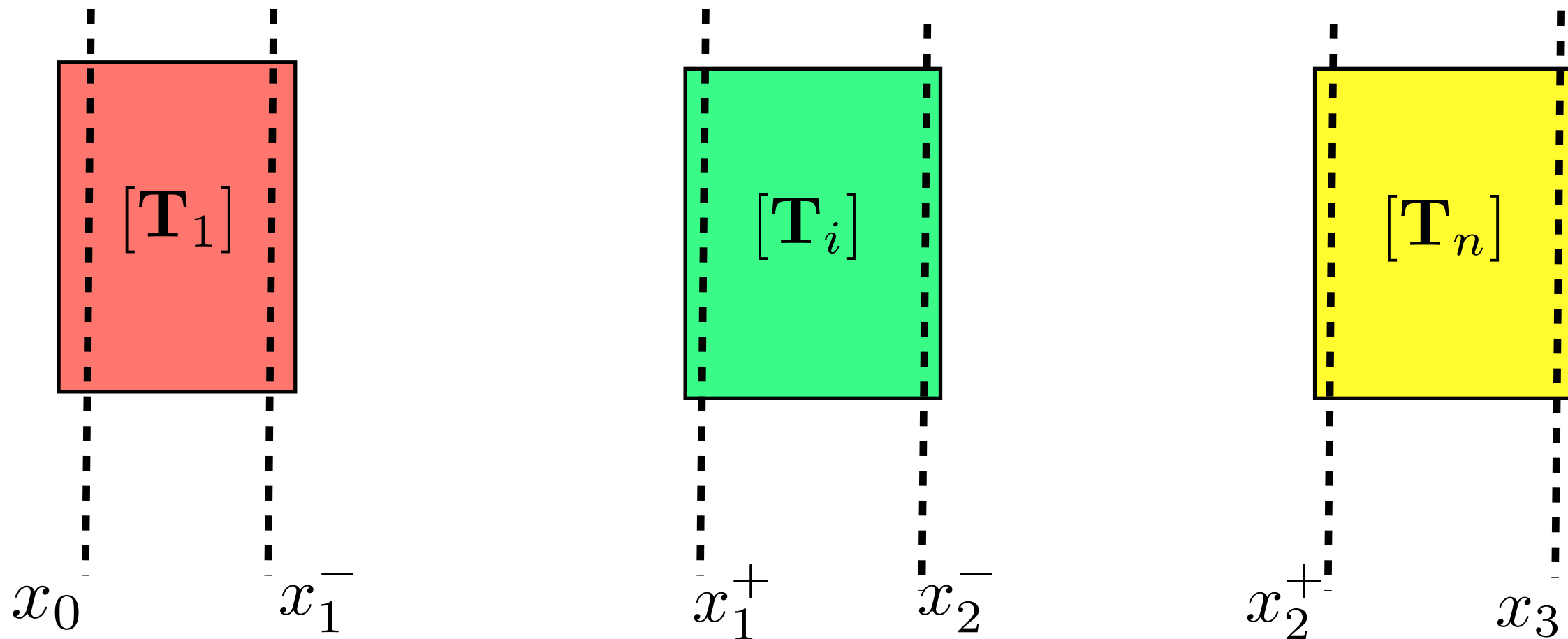
$$\mathbf{V}(z') = [\mathbf{T}(d, k_x, \text{medium properties})] \mathbf{V}(z)$$

Transfer matrix

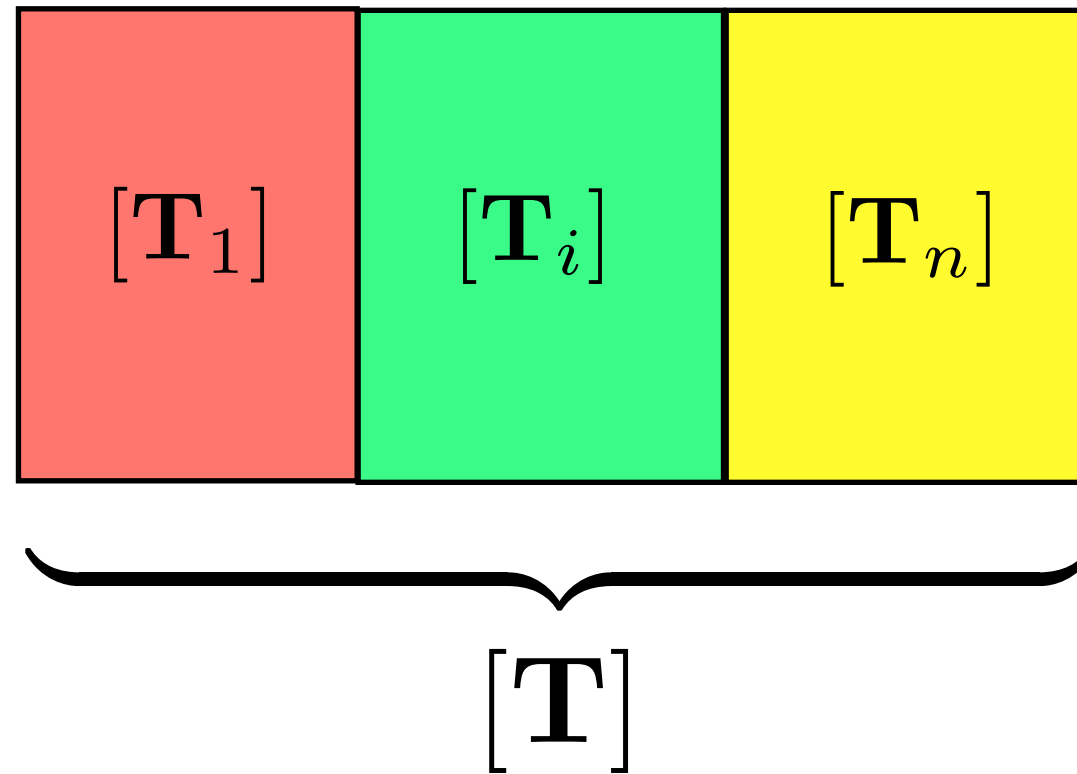


Interest of Transfert Matrices





If the media are of same type $V(x_i^-) = V(x_i^+)$



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Multiplication of transfert matrices

$$[T] = \prod_{i=1}^n [T_i]$$



- Z_{eq}
- k_{eq}
- ρ_{eq}
- K_{eq}



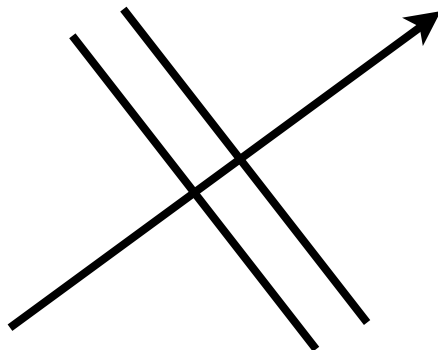
- Z_{eq}

- k_{eq}

- ρ_{eq}

- K_{eq}

k_x
 0
 k_z



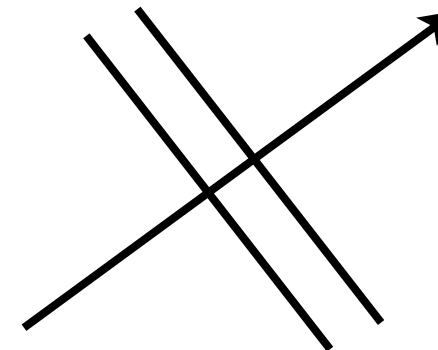


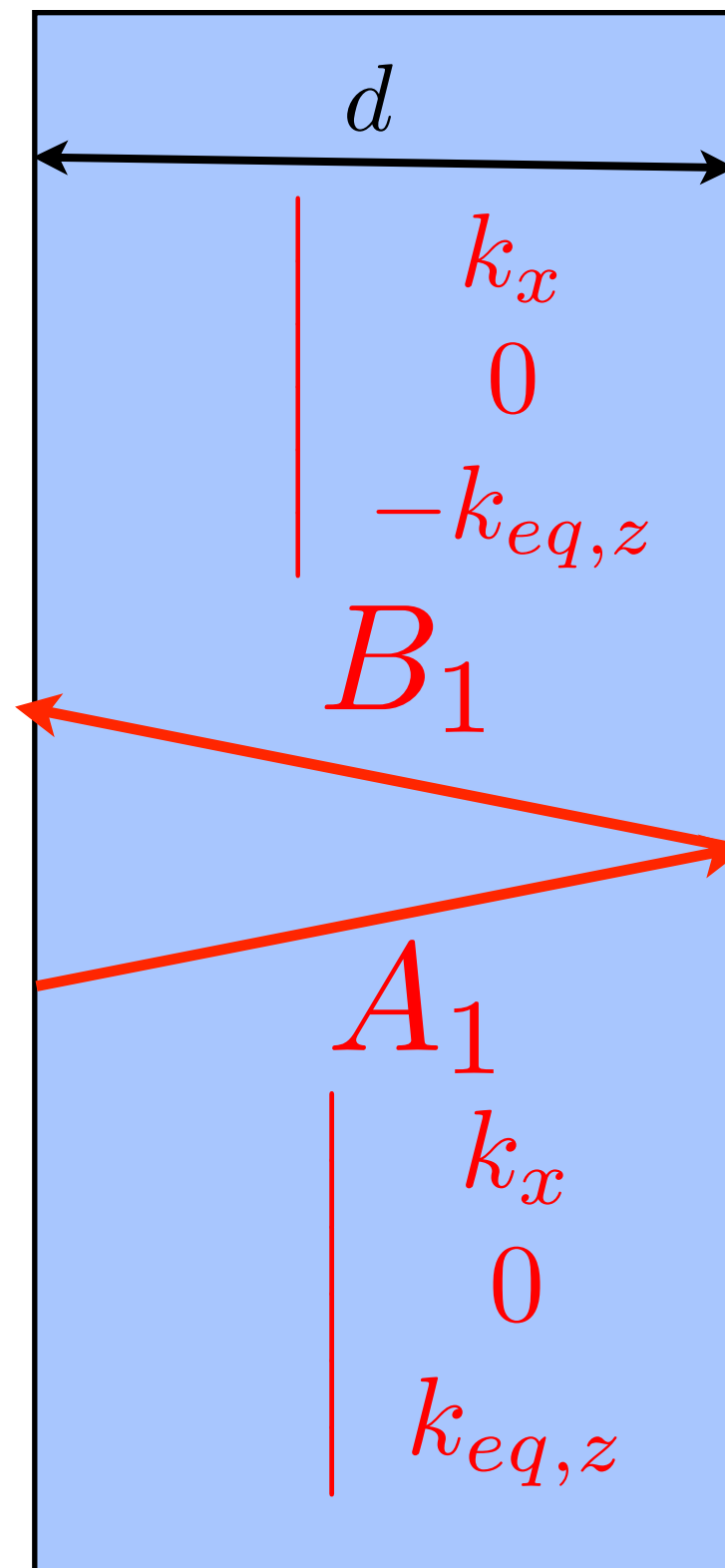
- Z_{eq}

- k_{eq}

- ρ_{eq}

- K_{eq}

$$\begin{vmatrix} k_x \\ 0 \\ k_z \end{vmatrix}$$




$$k_{eq}^2 = k_x^2 + k_{eq,z}^2$$

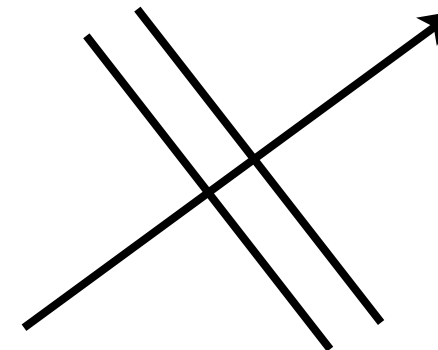
$$\mathbf{V}(0) = \begin{Bmatrix} P(0) \\ v(0) \end{Bmatrix}$$

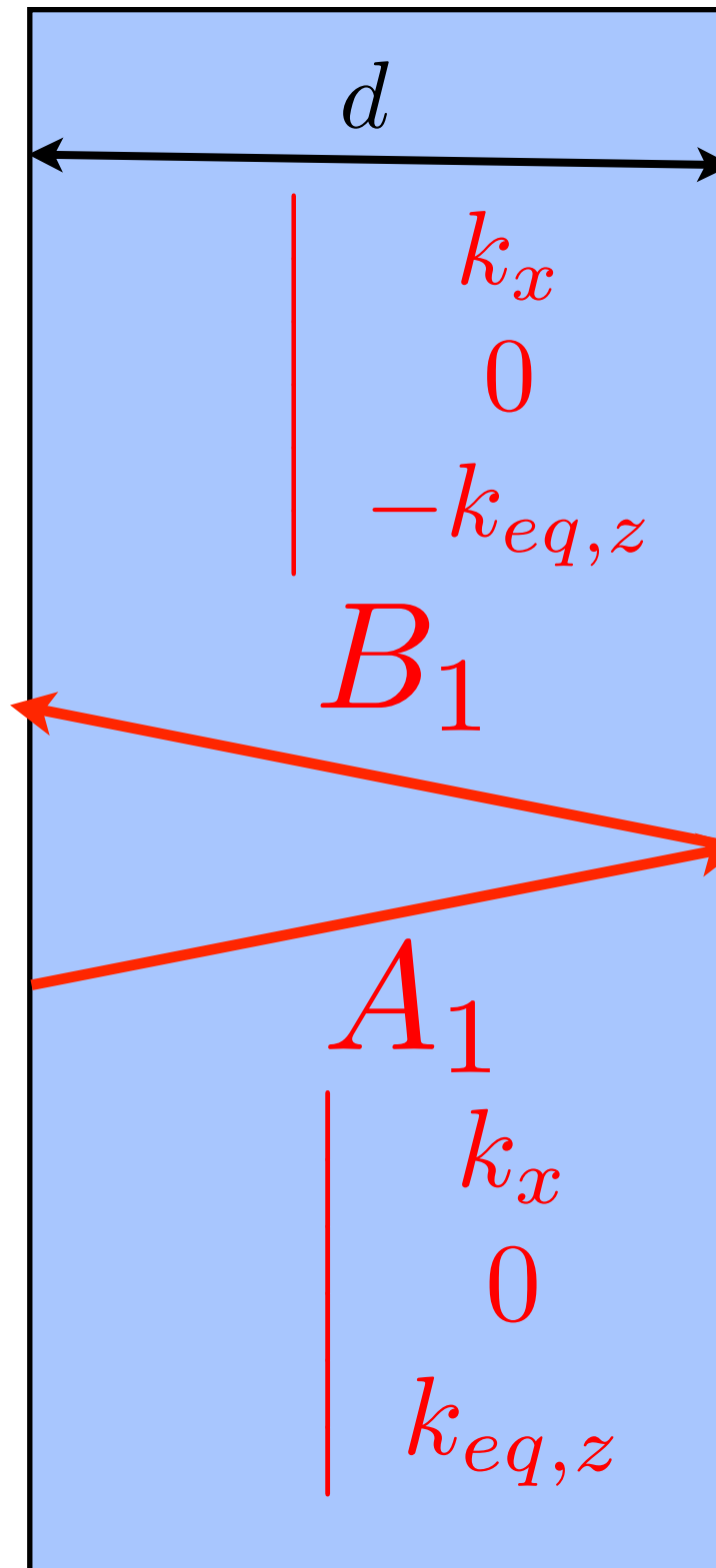
- Z_{eq}

- k_{eq}

- ρ_{eq}

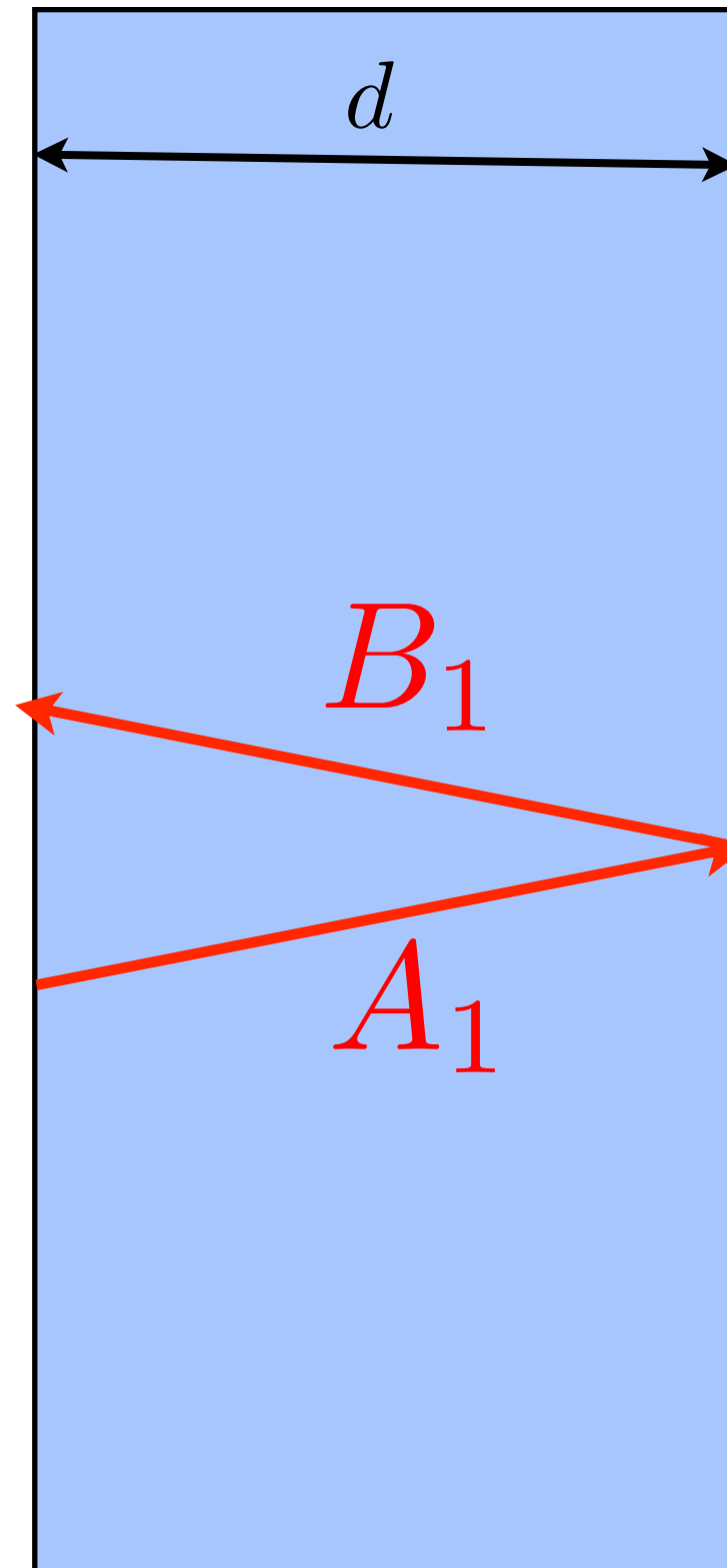
- K_{eq}

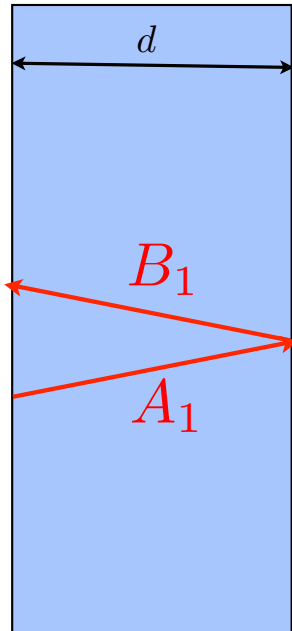
$$\begin{Bmatrix} k_x \\ 0 \\ k_z \end{Bmatrix}$$




$$\mathbf{V}(d) = \begin{Bmatrix} P(d) \\ v(d) \end{Bmatrix}$$

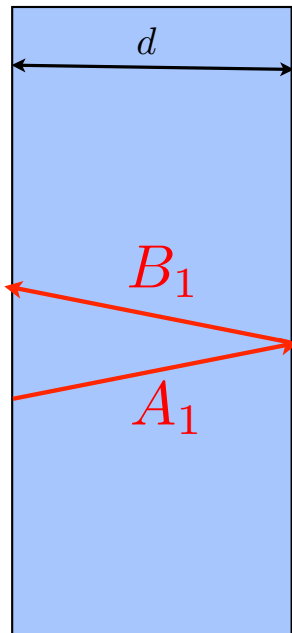
$$k_{eq}^2 = k_x^2 + k_{eq,z}^2$$





$$v(z) = A_1 e^{jk_{eq,z}z} + B_1 e^{-jk_{eq,z}z}$$

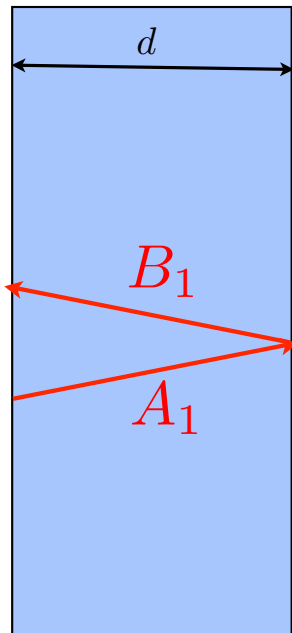
$$p(z) = -\frac{\omega \tilde{\rho}_{eq}}{k_{eq,z}} A_1 e^{jk_{eq,z}z} + \frac{\omega \tilde{\rho}_{eq}}{k_{eq,z}} B_1 e^{-jk_{eq,z}z}$$



$$v(z) = A_1 e^{jk_{eq,z}z} + B_1 e^{-jk_{eq,z}z}$$

$$p(z) = -\frac{\omega \tilde{\rho}_{eq}}{k_{eq,z}} A_1 e^{jk_{eq,z}z} + \frac{\omega \tilde{\rho}_{eq}}{k_{eq,z}} B_1 e^{-jk_{eq,z}z}$$

$$\begin{aligned} v(0) &= A_1 + B_1 \\ p(0) &= \frac{\omega \tilde{\rho}_{eq}}{k_{eq,z}} (-A_1 + B_1) \Rightarrow \begin{Bmatrix} A_1 \\ B_1 \end{Bmatrix} = \begin{bmatrix} \frac{1}{2} & \frac{k_{eq,z}}{2\omega \tilde{\rho}_{eq}} \\ \frac{1}{2} & -\frac{k_{eq,z}}{2\omega \tilde{\rho}_{eq}} \end{bmatrix} \begin{Bmatrix} v(0) \\ p(0) \end{Bmatrix} \end{aligned}$$



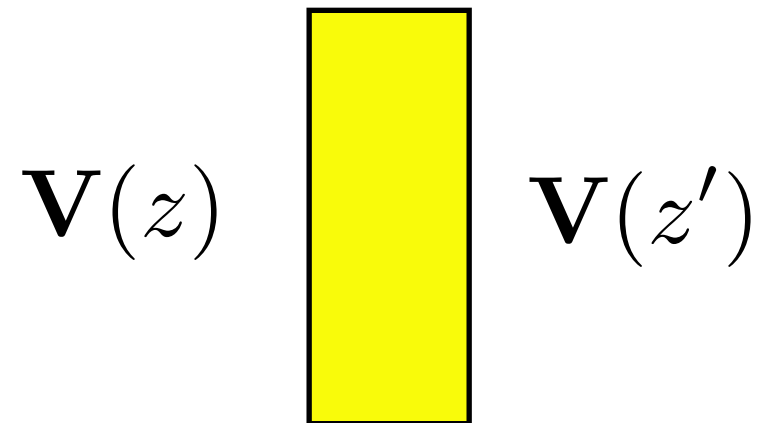
$$v(z) = A_1 e^{jk_{eq,z}z} + B_1 e^{-jk_{eq,z}z}$$

$$p(z) = -\frac{\omega \tilde{\rho}_{eq}}{k_{eq,z}} A_1 e^{jk_{eq,z}z} + \frac{\omega \tilde{\rho}_{eq}}{k_{eq,z}} B_1 e^{-jk_{eq,z}z}$$

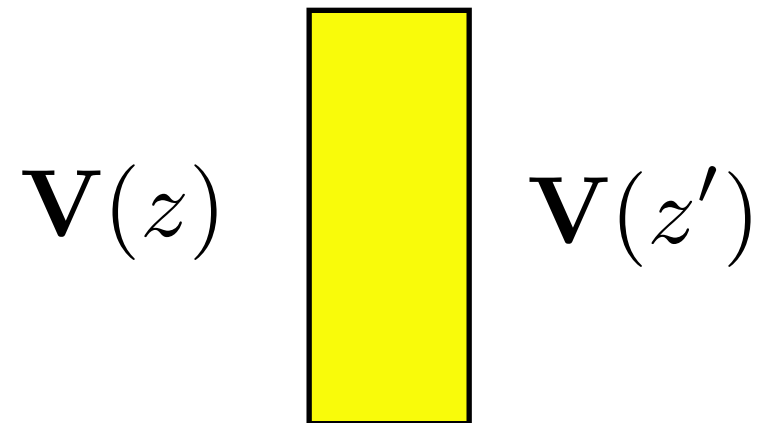
$$v(0) = A_1 + B_1$$

$$p(0) = \frac{\omega \tilde{\rho}_{eq}}{k_{eq,z}} (-A_1 + B_1) \Rightarrow \begin{Bmatrix} A_1 \\ B_1 \end{Bmatrix} = \begin{bmatrix} \frac{1}{2} & \frac{k_{eq,z}}{2\omega \tilde{\rho}_{eq}} \\ \frac{1}{2} & -\frac{k_{eq,z}}{2\omega \tilde{\rho}_{eq}} \end{bmatrix} \begin{Bmatrix} v(0) \\ p(0) \end{Bmatrix}$$

$$[\mathbf{T}] = \begin{bmatrix} \cos(k_{eq,z} d) & \frac{j\omega \tilde{\rho}_{eq}}{k_{eq,z}} \sin(k_{eq,z} d) \\ \frac{jk_{eq,z}}{\omega \tilde{\rho}_{eq}} \sin(k_{eq,z} d) & \cos(k_{eq,z} d) \end{bmatrix}$$



One key point:
Length of V = number of plane waves



One key point:

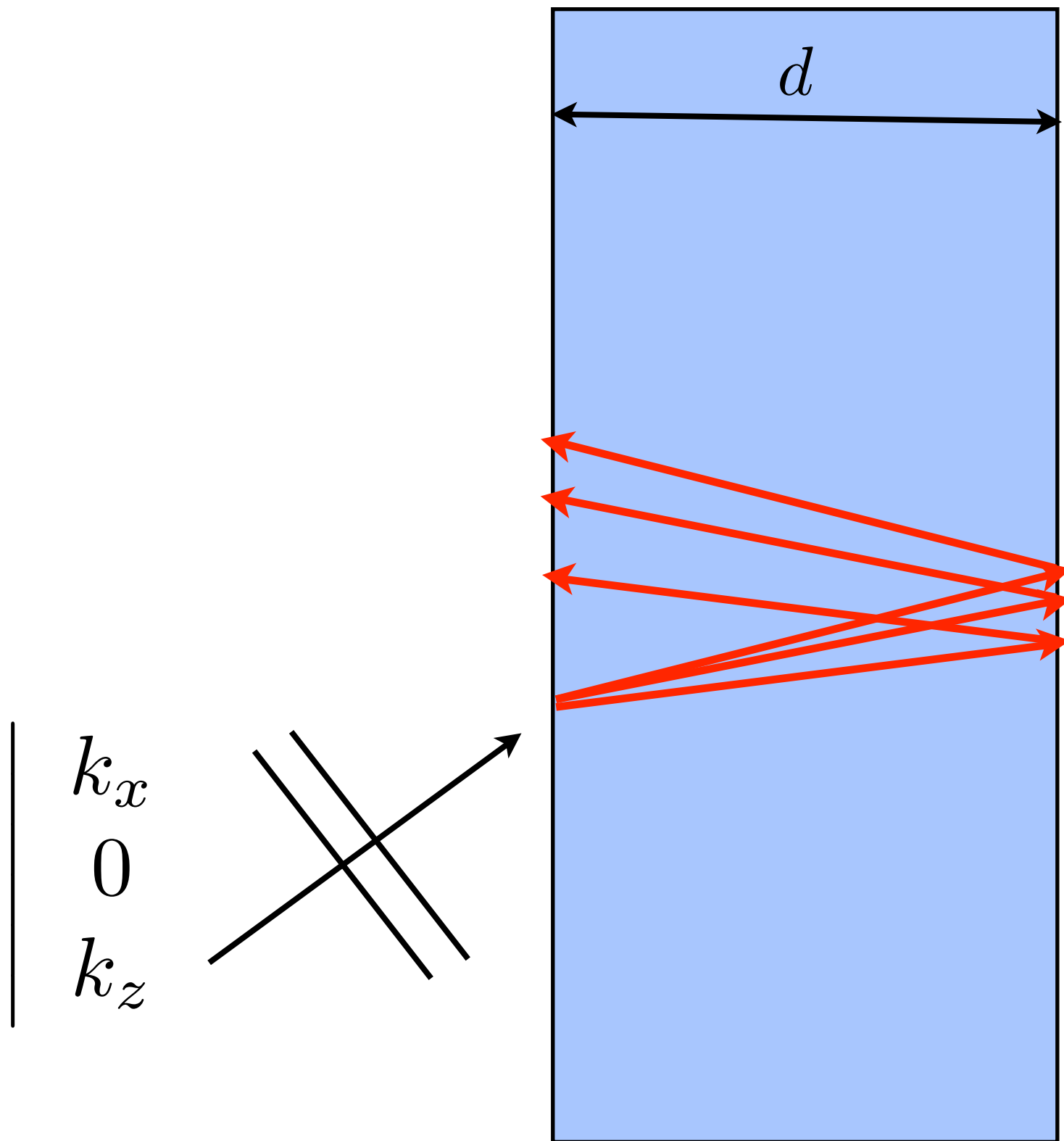
Length of V = number of plane waves

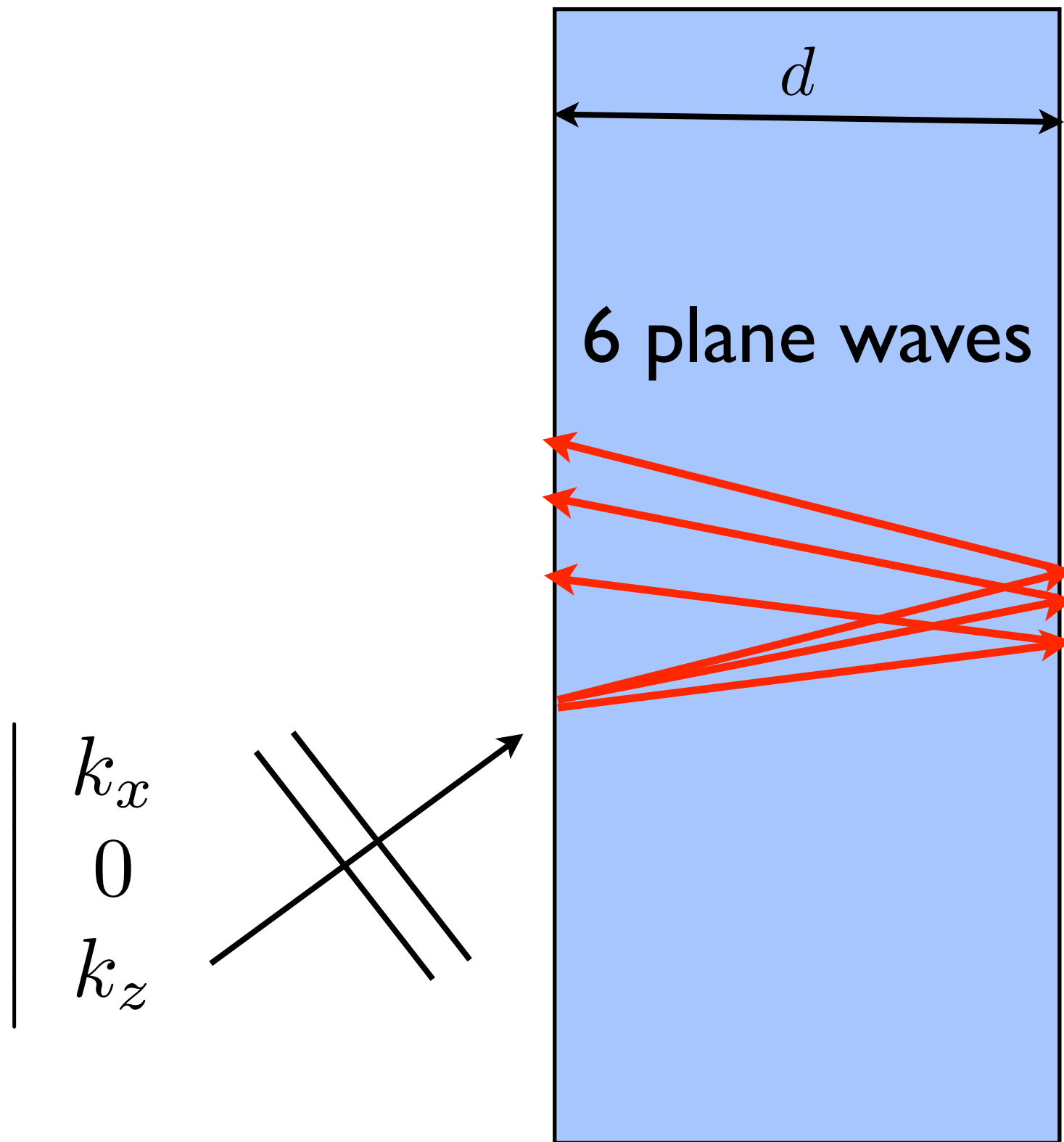
$$\mathbf{V}(z) = [\mathbf{A}(z)]\mathbf{q} \quad \mathbf{V}(z') = [\mathbf{A}(z')]\mathbf{q}$$

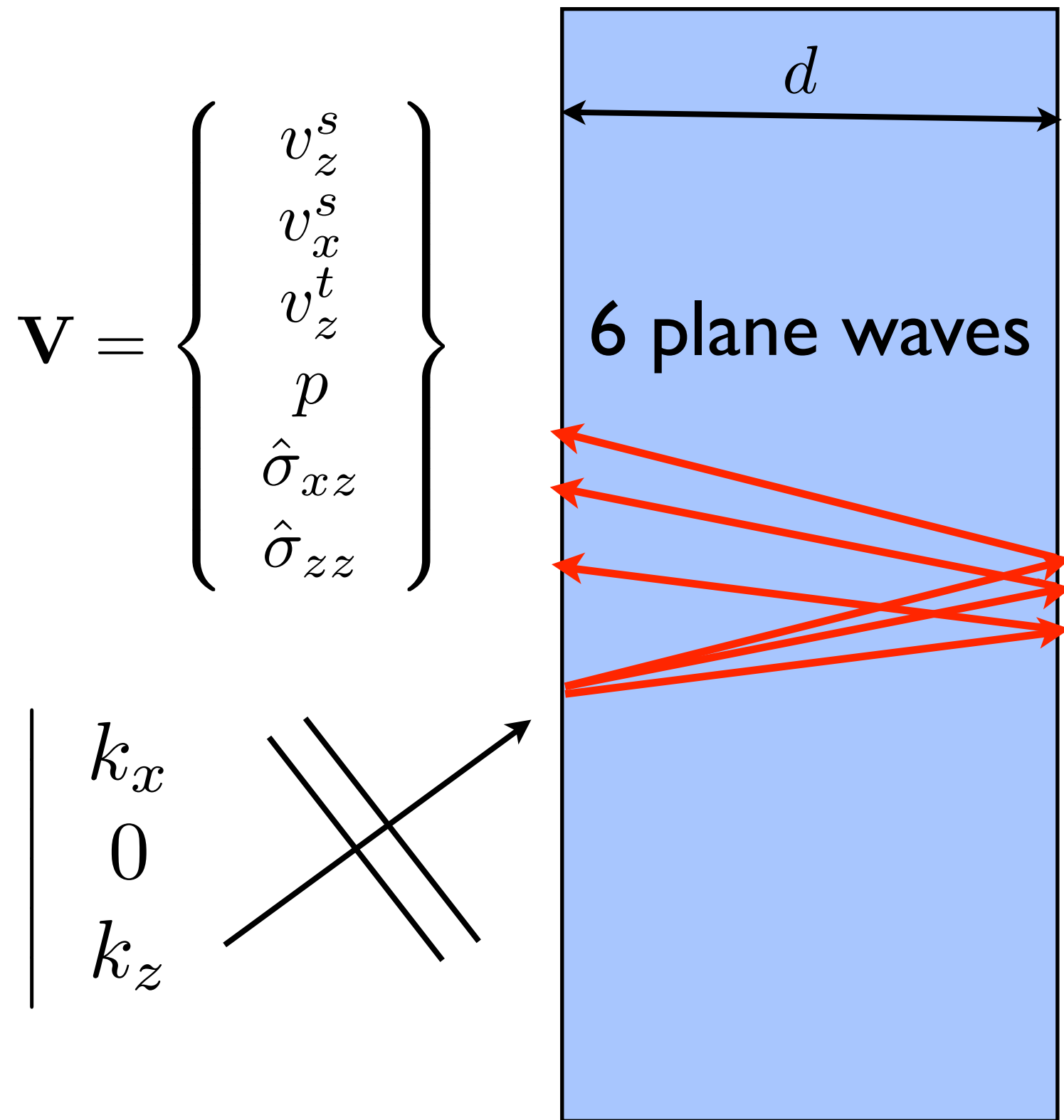
$$\mathbf{V}(z') = \underbrace{[\mathbf{A}(z')][\mathbf{A}(z)]^{-1}}_{[\mathbf{T}]} \mathbf{V}(z)$$

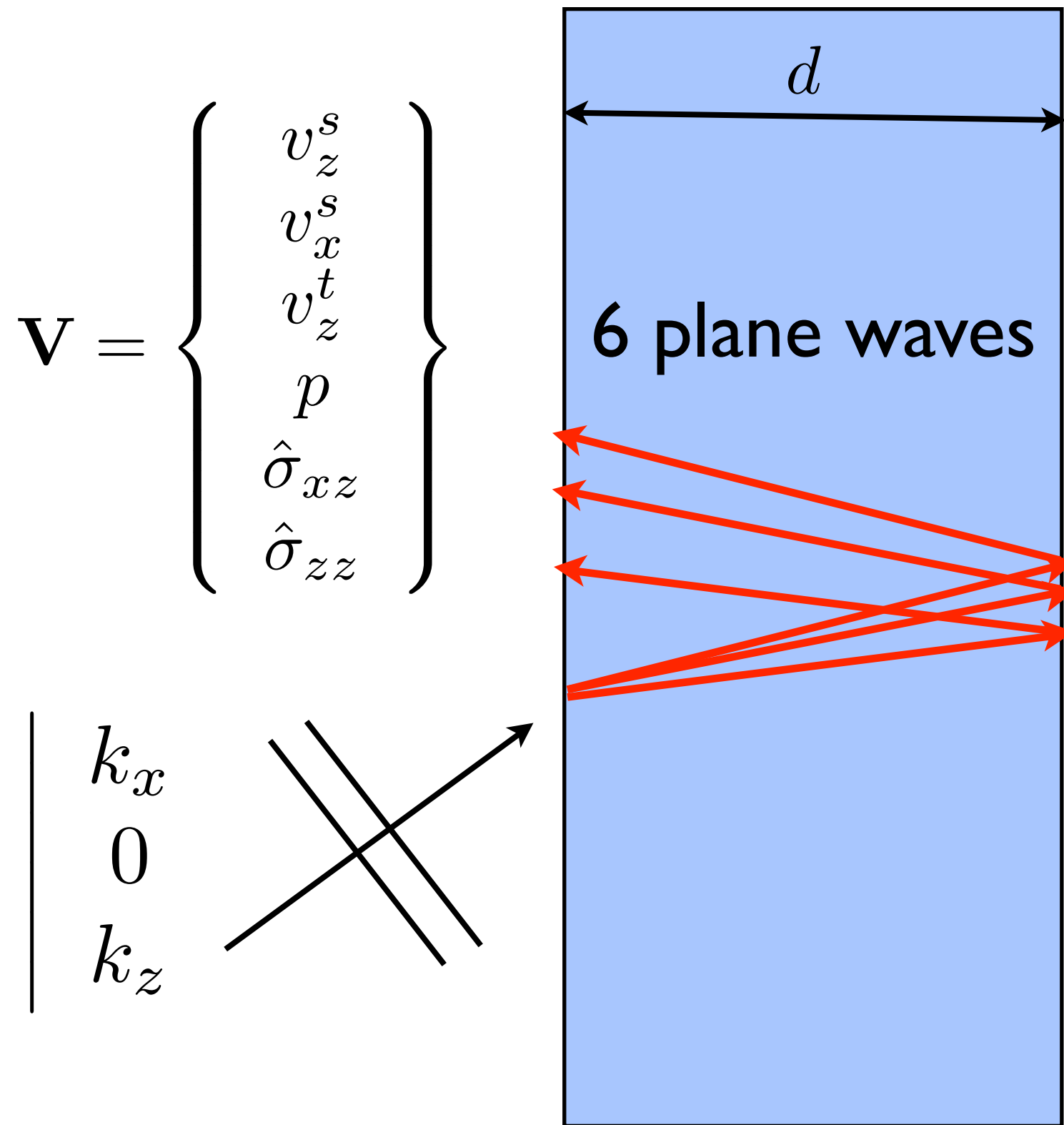
Analytically determined (if possible)





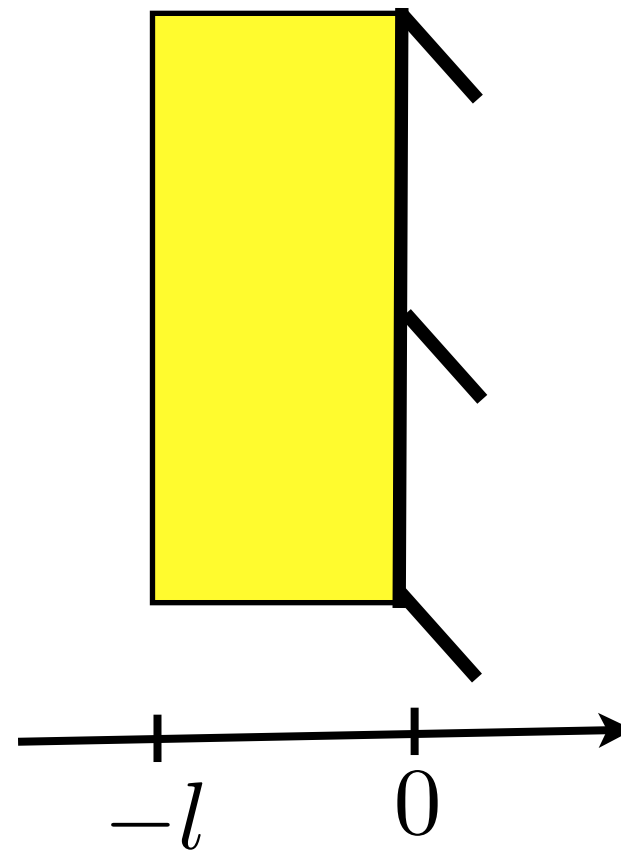






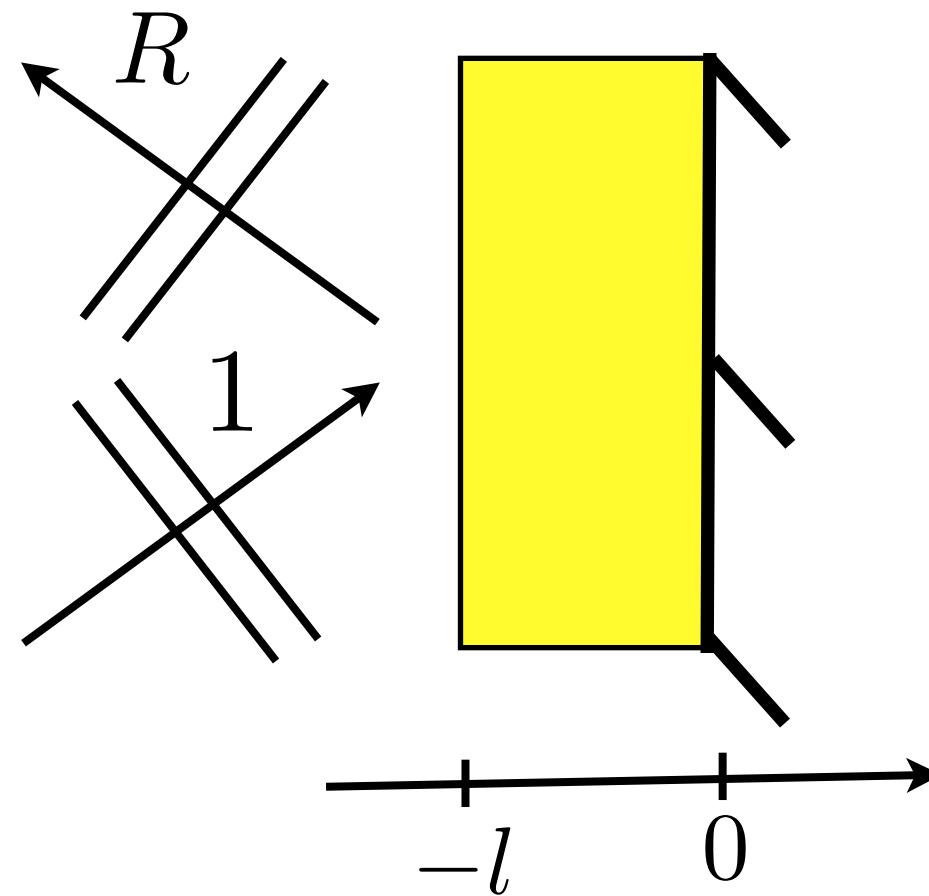
$$T_{ij} = \sum_{k=1}^3 \left(e_k + \frac{(-1)^{i+j}}{e_k} \right) r_i(k) c_j(k)$$

Absorption problem (fluids)



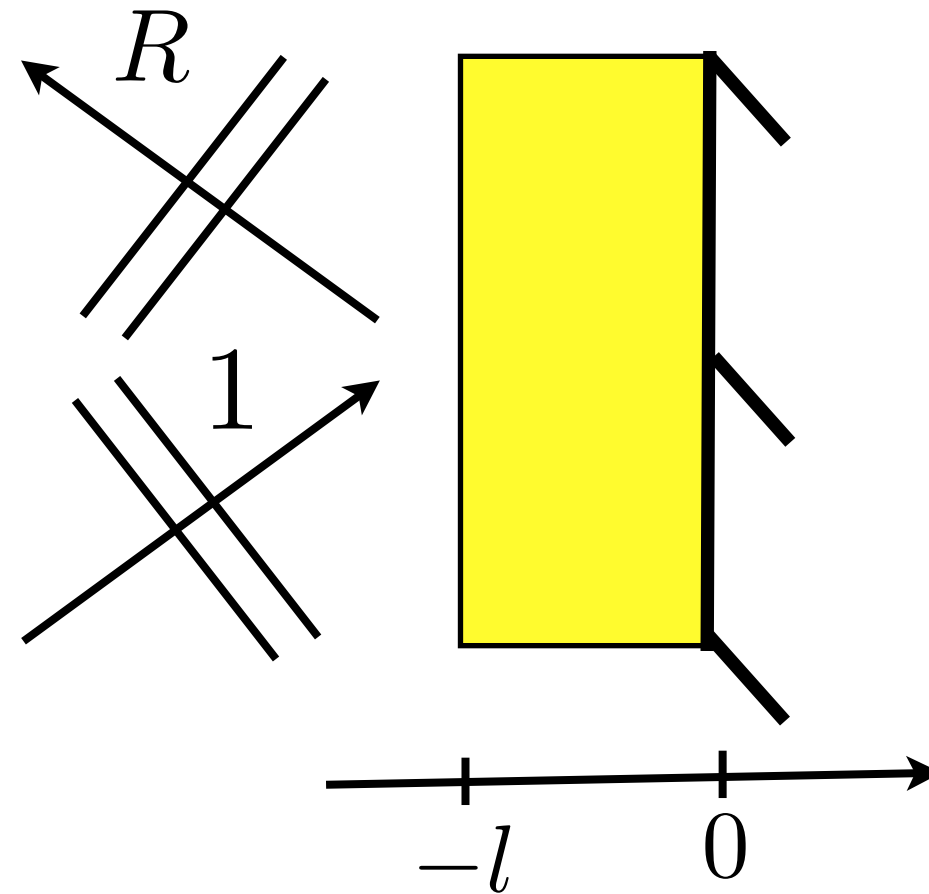
Rigid backing
Absorption pb

Absorption problem (fluids)



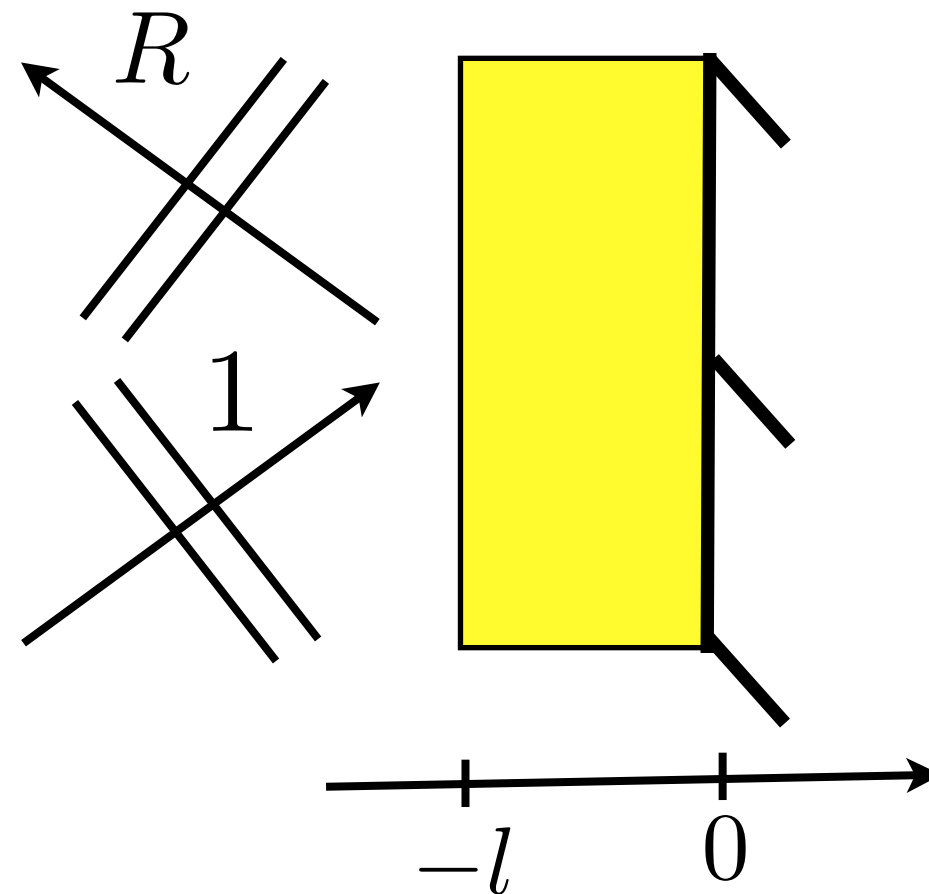
Rigid backing
Absorption pb

Absorption problem (fluids)



Rigid backing
Absorption pb

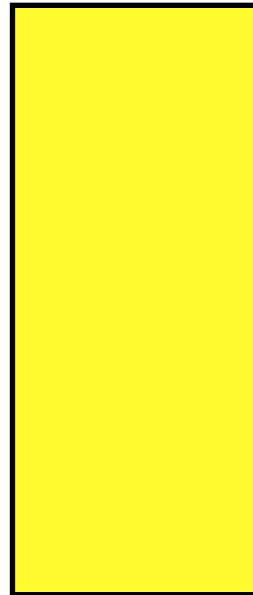
$$\begin{Bmatrix} v_z^t(-l) \\ p(-l) \end{Bmatrix} = \begin{bmatrix} T_{1,1} & T_{1,2} \\ T_{2,1} & T_{2,2} \end{bmatrix} \begin{Bmatrix} 0 \\ p(0) \end{Bmatrix}$$

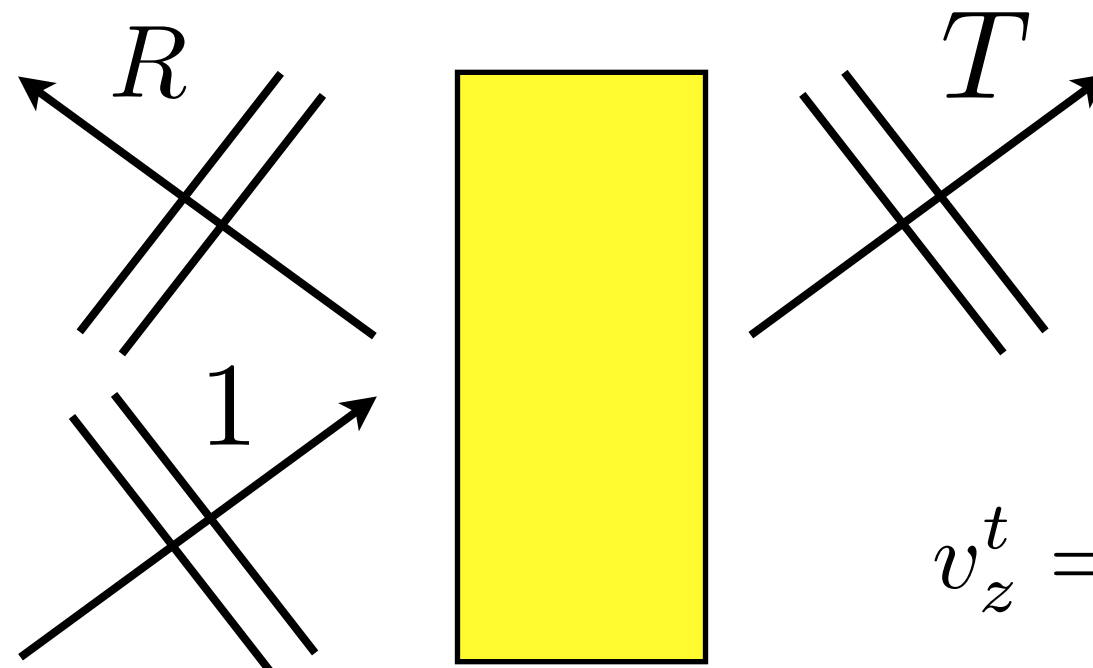


Rigid backing
Absorption pb

$$\begin{Bmatrix} v_z^t(-l) \\ p(-l) \end{Bmatrix} = \begin{bmatrix} T_{1,1} & T_{1,2} \\ T_{2,1} & T_{2,2} \end{bmatrix} \begin{Bmatrix} 0 \\ p(0) \end{Bmatrix}$$

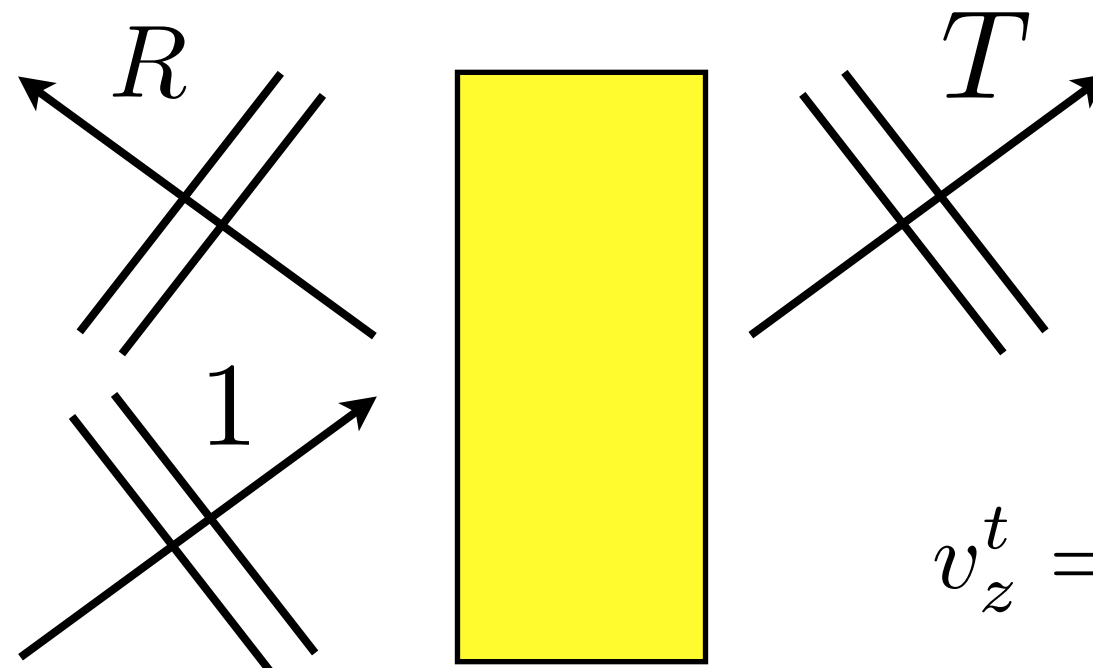
$$Z_s = \frac{p(-l)}{v(-l)} = \frac{T_{1,2}}{T_{2,2}}$$





$p(-l) = 1 + R$

$$v_z^a = \frac{k_0 \cos(\theta)}{\rho \omega} (R - 1)$$
$$v_z^t = -\frac{k_0 \cos(\theta)}{\rho \omega} p(0)$$

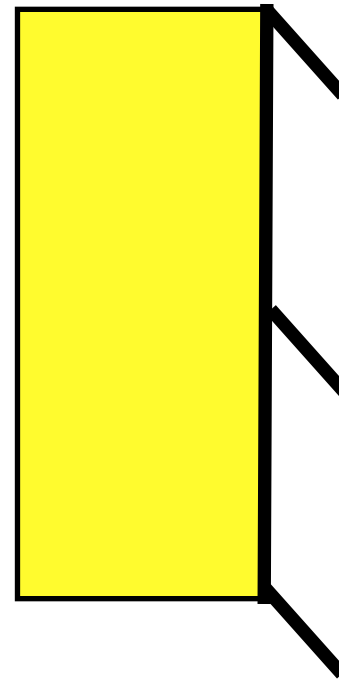


$$p(-l) = 1 + R$$

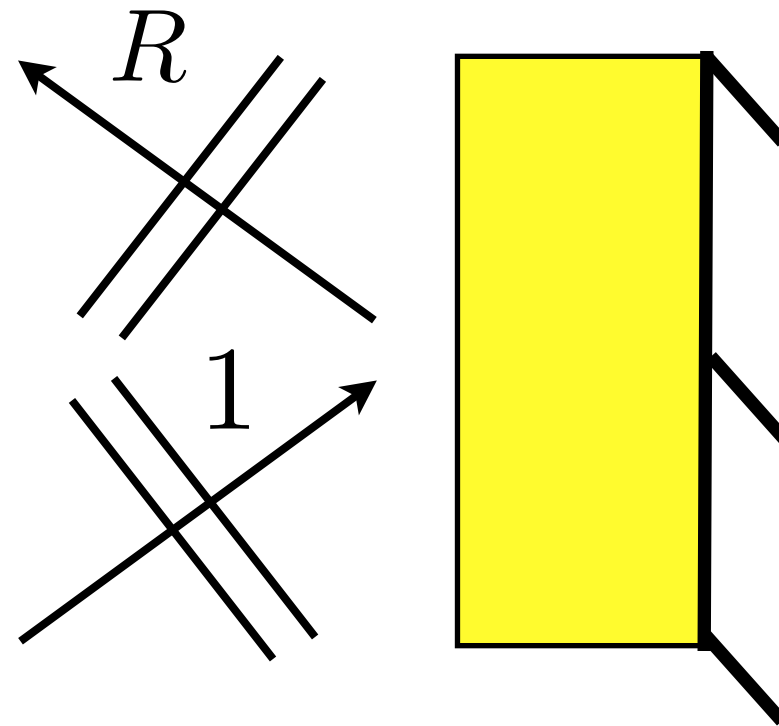
$$v_z^a = \frac{k_0 \cos(\theta)}{\rho \omega} (R - 1)$$

$$v_z^t = -\frac{k_0 \cos(\theta)}{\rho \omega} p(0)$$

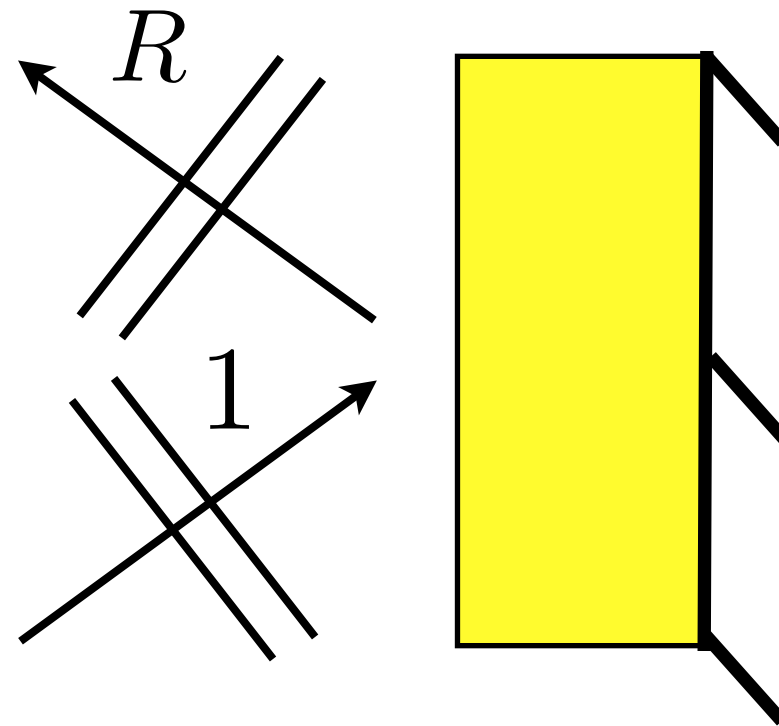
$$\begin{Bmatrix} R - 1 \\ R + 1 \end{Bmatrix} = \begin{bmatrix} \frac{T_{1,1} \rho \omega}{k_0 \cos(\theta)} & \frac{T_{1,2} \rho \omega}{k_0 \cos(\theta)} \\ T_{2,1} & T_{2,2} \end{bmatrix} \begin{Bmatrix} -\frac{k_0 \cos(\theta)}{\rho \omega} \\ 1 \end{Bmatrix} T$$



Rigid backing
Absorption pb

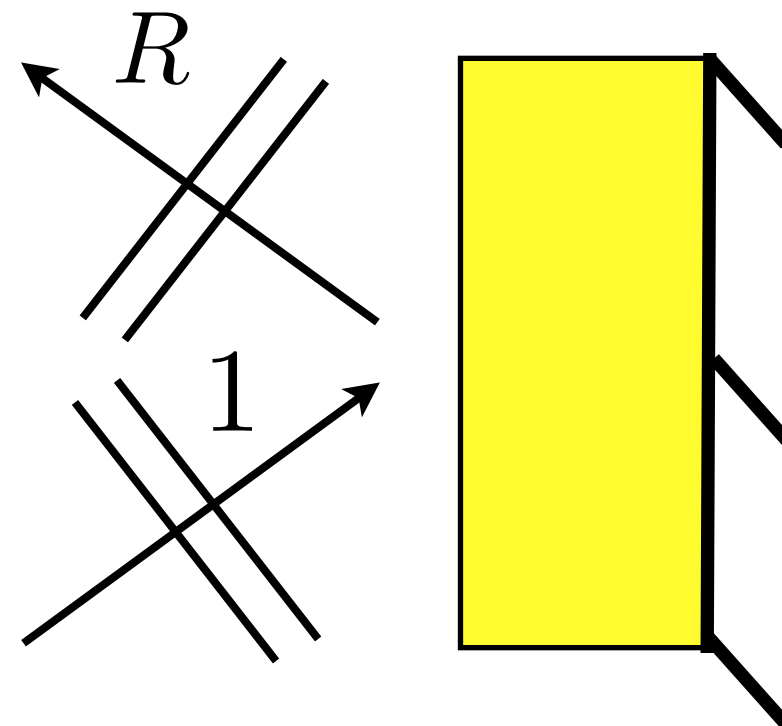


Rigid backing
Absorption pb



Rigid backing
Absorption pb

$$\mathbf{V}(0) = \left\{ \begin{array}{l} v_z^s = 0 \\ v_x^s = 0 \\ v_z^t = 0 \\ p \\ \hat{\sigma}_{xz} \\ \hat{\sigma}_{zz} \end{array} \right\}$$



Rigid backing
Absorption pb

$$\mathbf{V}(-l) = \left\{ \begin{array}{l} v_z^s(-l) \\ v_x^s(-l) \\ v_z^t(-l) \\ p(-l) \\ \hat{\sigma}_{xz} = 0 \\ \hat{\sigma}_{zz} = 0 \end{array} \right\}$$

$$\mathbf{V}(0) = \left\{ \begin{array}{l} v_z^s = 0 \\ v_x^s = 0 \\ v_z^t = 0 \\ p \\ \hat{\sigma}_{xz} \\ \hat{\sigma}_{zz} \end{array} \right\}$$

$$\begin{pmatrix} v_z^s(-l) \\ v_x^s(-l) \\ v_z^t(-l) \\ p(-l) \\ \hat{\sigma}_{xz} = 0 \\ \hat{\sigma}_{zz} = 0 \end{pmatrix} = \begin{bmatrix} T_{1,1} & T_{1,2} & T_{1,3} & T_{1,4} & T_{1,5} & T_{1,6} \\ T_{2,1} & T_{2,2} & T_{2,3} & T_{2,4} & T_{2,5} & T_{2,6} \\ T_{3,1} & T_{3,2} & T_{3,3} & T_{3,4} & T_{3,5} & T_{3,6} \\ T_{4,1} & T_{4,2} & T_{4,3} & T_{4,4} & T_{4,5} & T_{4,6} \\ T_{5,1} & T_{5,2} & T_{5,3} & T_{5,4} & T_{5,5} & T_{5,6} \\ T_{6,1} & T_{6,2} & T_{6,3} & T_{6,4} & T_{6,5} & T_{6,6} \end{bmatrix} \begin{pmatrix} 0 \\ 0 \\ 0 \\ p(0) \\ \hat{\sigma}_{xz}(0) \\ \hat{\sigma}_{zz}(0) \end{pmatrix}$$

$$\left\{ \begin{array}{l} v_z^s(-l) \\ v_x^s(-l) \\ v_z^t(-l) \\ p(-l) \\ \hat{\sigma}_{xz} = 0 \\ \hat{\sigma}_{zz} = 0 \end{array} \right\} = \left[\begin{array}{cccccc} T_{1,1} & T_{1,2} & T_{1,3} & T_{1,4} & T_{1,5} & T_{1,6} \\ T_{2,1} & T_{2,2} & T_{2,3} & T_{2,4} & T_{2,5} & T_{2,6} \\ T_{3,1} & T_{3,2} & T_{3,3} & T_{3,4} & T_{3,5} & T_{3,6} \\ T_{4,1} & T_{4,2} & T_{4,3} & T_{4,4} & T_{4,5} & T_{4,6} \\ T_{5,1} & T_{5,2} & T_{5,3} & T_{5,4} & T_{5,5} & T_{5,6} \\ T_{6,1} & T_{6,2} & T_{6,3} & T_{6,4} & T_{6,5} & T_{6,6} \end{array} \right] \left\{ \begin{array}{l} \text{---} 0 \text{---} \\ \text{---} 0 \text{---} \\ \text{---} 0 \text{---} \\ p(0) \\ \hat{\sigma}_{xz}(0) \\ \hat{\sigma}_{zz}(0) \end{array} \right\}$$

$$\left\{ \begin{array}{l} v_z^s(-l) \\ v_x^s(-l) \\ v_z^t(-l) \\ p(-l) \\ \hat{\sigma}_{xz} = 0 \\ \hat{\sigma}_{zz} = 0 \end{array} \right\} = \left[\begin{array}{cccccc} T_{1,1} & T_{1,2} & T_{1,3} & T_{1,4} & T_{1,5} & T_{1,6} \\ T_{2,1} & T_{2,2} & T_{2,3} & T_{2,4} & T_{2,5} & T_{2,6} \\ T_{3,1} & T_{3,2} & T_{3,3} & T_{3,4} & T_{3,5} & T_{3,6} \\ T_{4,1} & T_{4,2} & T_{4,3} & T_{4,4} & T_{4,5} & T_{4,6} \\ T_{5,1} & T_{5,2} & T_{5,3} & T_{5,4} & T_{5,5} & T_{5,6} \\ T_{6,1} & T_{6,2} & T_{6,3} & T_{6,4} & T_{6,5} & T_{6,6} \end{array} \right] \left\{ \begin{array}{l} \cancel{0} \\ \cancel{0} \\ \cancel{0} \\ p(0) \\ \hat{\sigma}_{xz}(0) \\ \hat{\sigma}_{zz}(0) \end{array} \right\}$$

$$\left[\begin{array}{cc} T_{5,5} & T_{5,6} \\ T_{6,5} & T_{6,6} \end{array} \right] \left\{ \begin{array}{l} \hat{\sigma}_{xz}(0) \\ \hat{\sigma}_{zz}(0) \end{array} \right\} = - \left\{ \begin{array}{l} T_{5,4} \\ T_{6,4} \end{array} \right\} p(0)$$

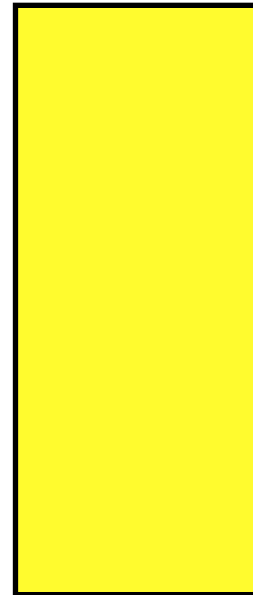
$$\left\{ \begin{array}{l} v_z^s(-l) \\ v_x^s(-l) \\ v_z^t(-l) \\ p(-l) \\ \hat{\sigma}_{xz} = 0 \\ \hat{\sigma}_{zz} = 0 \end{array} \right\} = \left[\begin{array}{cccccc} T_{1,1} & T_{1,2} & T_{1,3} & T_{1,4} & T_{1,5} & T_{1,6} \\ T_{2,1} & T_{2,2} & T_{2,3} & T_{2,4} & T_{2,5} & T_{2,6} \\ T_{3,1} & T_{3,2} & T_{3,3} & T_{3,4} & T_{3,5} & T_{3,6} \\ T_{4,1} & T_{4,2} & T_{4,3} & T_{4,4} & T_{4,5} & T_{4,6} \\ T_{5,1} & T_{5,2} & T_{5,3} & T_{5,4} & T_{5,5} & T_{5,6} \\ T_{6,1} & T_{6,2} & T_{6,3} & T_{6,4} & T_{6,5} & T_{6,6} \end{array} \right] \left\{ \begin{array}{l} \cancel{0} \\ \cancel{0} \\ \cancel{0} \\ p(0) \\ \hat{\sigma}_{xz}(0) \\ \hat{\sigma}_{zz}(0) \end{array} \right\}$$

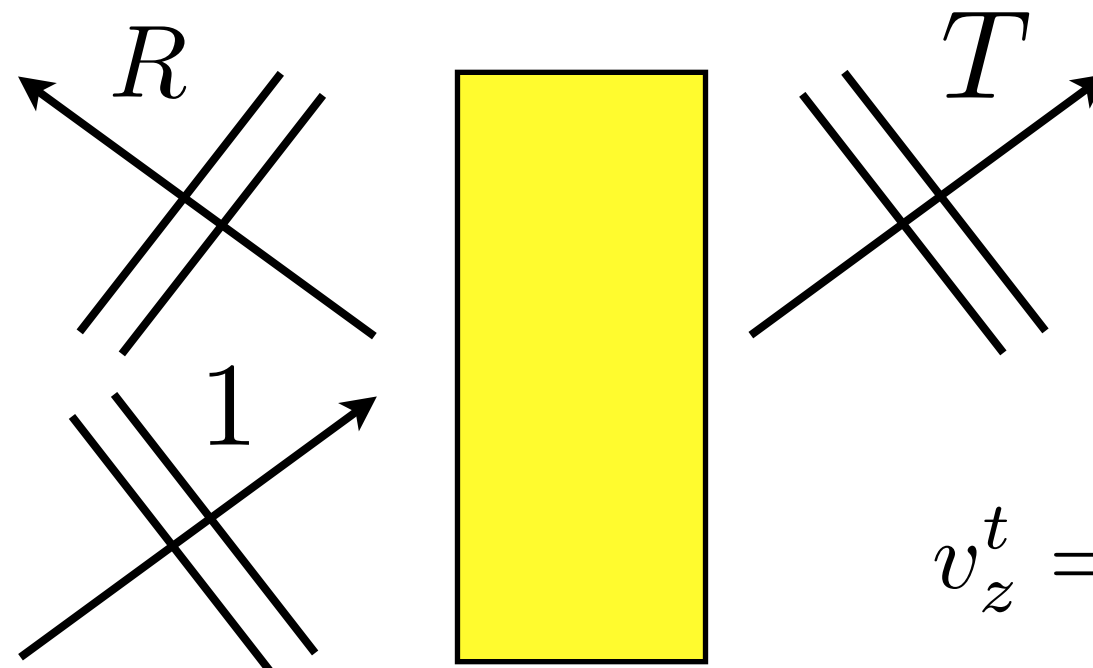
$$\hat{\sigma}_{xz}(0) = \alpha p(0), \quad \hat{\sigma}_{zz}(0) = \beta p(0)$$

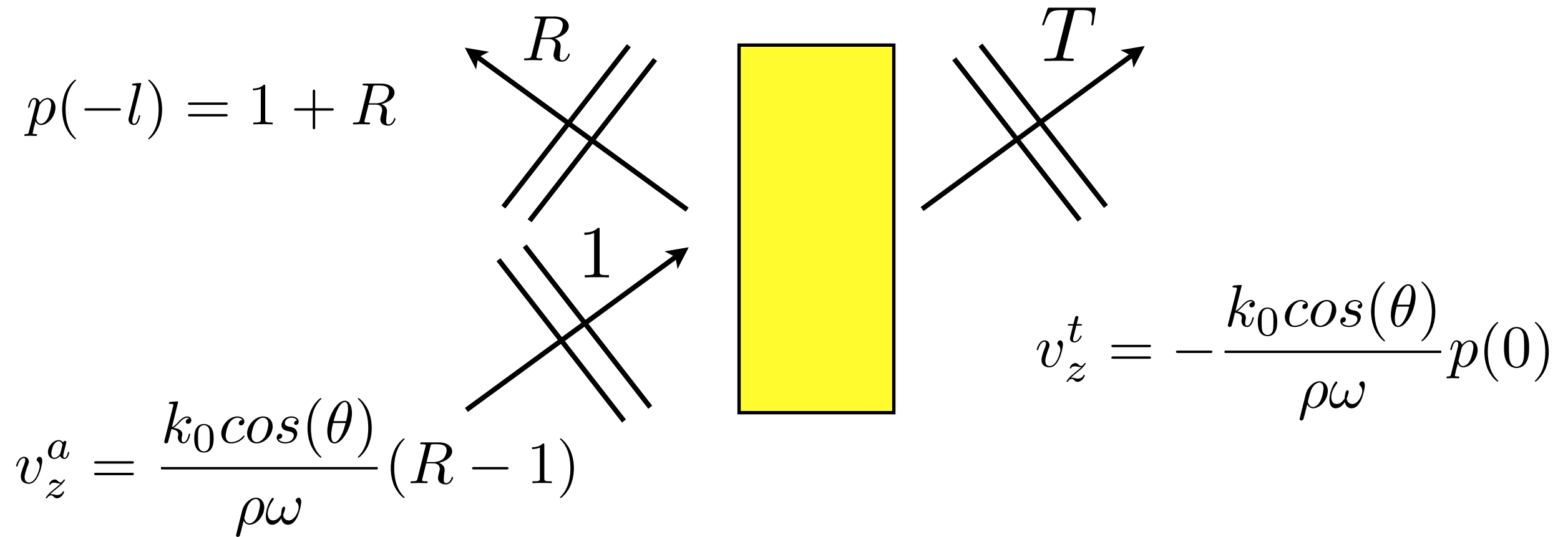
$$\left\{ \begin{array}{l} v_z^s(-l) \\ v_x^s(-l) \\ v_z^t(-l) \\ p(-l) \\ \hat{\sigma}_{xz} = 0 \\ \hat{\sigma}_{zz} = 0 \end{array} \right\} = \left[\begin{array}{cccccc} T_{1,1} & T_{1,2} & T_{1,3} & T_{1,4} & T_{1,5} & T_{1,6} \\ T_{2,1} & T_{2,2} & T_{2,3} & T_{2,4} & T_{2,5} & T_{2,6} \\ T_{3,1} & T_{3,2} & T_{3,3} & T_{3,4} & T_{3,5} & T_{3,6} \\ T_{4,1} & T_{4,2} & T_{4,3} & T_{4,4} & T_{4,5} & T_{4,6} \\ T_{5,1} & T_{5,2} & T_{5,3} & T_{5,4} & T_{5,5} & T_{5,6} \\ T_{6,1} & T_{6,2} & T_{6,3} & T_{6,4} & T_{6,5} & T_{6,6} \end{array} \right] \left\{ \begin{array}{l} \cancel{0} \\ \cancel{0} \\ \cancel{0} \\ p(0) \\ \hat{\sigma}_{xz}(0) \\ \hat{\sigma}_{zz}(0) \end{array} \right\}$$

$$\hat{\sigma}_{xz}(0) = \alpha p(0), \quad \hat{\sigma}_{zz}(0) = \beta p(0)$$

$$Z_s = \frac{p(-l)}{v(-l)} = \frac{T_{4,4} + \alpha T_{4,5} + \beta T_{4,6}}{T_{3,4} + \alpha T_{3,5} + \beta T_{3,6}}$$




$$p(-l) = 1 + R$$
$$v_z^a = \frac{k_0 \cos(\theta)}{\rho \omega} (R - 1)$$
$$v_z^t = - \frac{k_0 \cos(\theta)}{\rho \omega} p(0)$$



$$\mathbf{V}(-l) = \left\{ \begin{array}{c} v_z^s(-l) \\ v_x^s(-l) \\ v_z^t(-l) \\ p(-l) \\ \hat{\sigma}_{xz} = 0 \\ \hat{\sigma}_{zz} = 0 \end{array} \right\}$$

$$\mathbf{V}(0) = \left\{ \begin{array}{c} v_z^s(0) \\ v_x^s(0) \\ v_z^t(0) \\ p(0) \\ \hat{\sigma}_{xz} = 0 \\ \hat{\sigma}_{zz} = 0 \end{array} \right\}$$

$$\begin{pmatrix} v_z^s(-l) \\ v_x^s(-l) \\ v_z^t(-l) \\ p(-l) \\ \hat{\sigma}_{xz} = 0 \\ \hat{\sigma}_{zz} = 0 \end{pmatrix} \begin{bmatrix} T_{1,1} & T_{1,2} & T_{1,3} & T_{1,4} & T_{1,5} & T_{1,6} \\ T_{2,1} & T_{2,2} & T_{2,3} & T_{2,4} & T_{2,5} & T_{2,6} \\ T_{3,1} & T_{3,2} & T_{3,3} & T_{3,4} & T_{3,5} & T_{3,6} \\ T_{4,1} & T_{4,2} & T_{4,3} & T_{4,4} & T_{4,5} & T_{4,6} \\ T_{5,1} & T_{5,2} & T_{5,3} & T_{5,4} & T_{5,5} & T_{5,6} \\ T_{6,1} & T_{6,2} & T_{6,3} & T_{6,4} & T_{6,5} & T_{6,6} \end{bmatrix} \begin{pmatrix} v_z^s(0) \\ v_x^s(0) \\ v_z^t(0) \\ p(0) \\ \hat{\sigma}_{xz} = 0 \\ \hat{\sigma}_{zz} = 0 \end{pmatrix}$$

$$\begin{pmatrix} v_z^s(-l) \\ v_x^s(-l) \\ v_z^t(-l) \\ p(-l) \\ \hat{\sigma}_{xz} = 0 \\ \hat{\sigma}_{zz} = 0 \end{pmatrix} \left[\begin{array}{cc|cc|cc} T_{1,1} & T_{1,2} & T_{1,3} & T_{1,4} & T_{1,5} & T_{1,6} \\ T_{2,1} & T_{2,2} & T_{2,3} & T_{2,4} & T_{2,5} & T_{2,6} \\ T_{3,1} & T_{3,2} & T_{3,3} & T_{3,4} & T_{3,5} & T_{3,6} \\ T_{4,1} & T_{4,2} & T_{4,3} & T_{4,4} & T_{4,5} & T_{4,6} \\ T_{5,1} & T_{5,2} & T_{5,3} & T_{5,4} & T_{5,5} & T_{5,6} \\ T_{6,1} & T_{6,2} & T_{6,3} & T_{6,4} & T_{6,5} & T_{6,6} \end{array} \right] \begin{pmatrix} v_z^s(0) \\ v_x^s(0) \\ v_z^t(0) / \frac{k_0 \cos(\theta)}{\rho\omega} \\ p(0) \\ \hat{\sigma}_{xz} = 0 \\ \hat{\sigma}_{zz} = 0 \end{pmatrix}$$

$\swarrow \frac{k_0 \cos(\theta)}{\rho\omega}$

$\begin{matrix} + \\ \times \frac{k_0 \cos(\theta)}{\rho\omega} \end{matrix}$

$$\begin{pmatrix} v_z^s(-l) \\ v_x^s(-l) \\ v_z^t(-l) \\ p(-l) \\ \hat{\sigma}_{xz} = 0 \\ \hat{\sigma}_{zz} = 0 \end{pmatrix} \begin{bmatrix} T_{1,1} & T_{1,2} & T_{1,3} & T_{1,4} & T_{1,5} & T_{1,6} \\ T_{2,1} & T_{2,2} & T_{2,3} & T_{2,4} & T_{2,5} & T_{2,6} \\ T_{3,1} & T_{3,2} & T_{3,3} & T_{3,4} & T_{3,5} & T_{3,6} \\ T_{4,1} & T_{4,2} & T_{4,3} & T_{4,4} & T_{4,5} & T_{4,6} \\ T_{5,1} & T_{5,2} & T_{5,3} & T_{5,4} & T_{5,5} & T_{5,6} \\ T_{6,1} & T_{6,2} & T_{6,3} & T_{6,4} & T_{6,5} & T_{6,6} \end{bmatrix} \begin{pmatrix} v_z^s(0) \\ v_x^s(0) \\ v_z^t(0) / \frac{k_0 \cos(\theta)}{\rho\omega} \\ p(0) \\ \hat{\sigma}_{xz} = 0 \\ \hat{\sigma}_{zz} = 0 \end{pmatrix}$$

$\swarrow \frac{k_0 \cos(\theta)}{\rho\omega}$

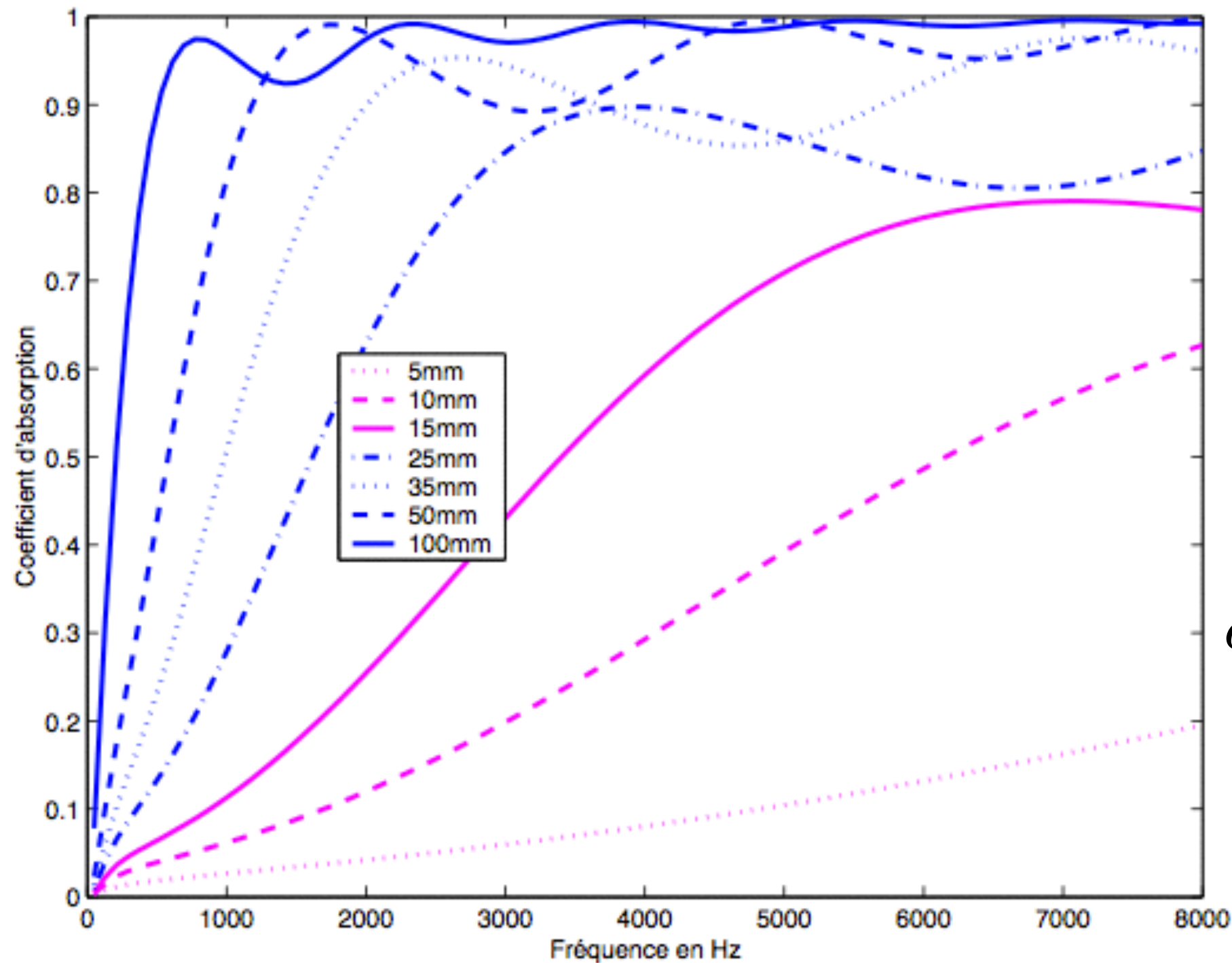
 $+ \times \frac{k_0 \cos(\theta)}{\rho\omega}$
 $\searrow \frac{k_0 \cos(\theta)}{\rho\omega}$

$$\begin{pmatrix} R - 1 \\ R + 1 \\ 0 \\ 0 \end{pmatrix} \begin{bmatrix} T'_{3,1} & T'_{3,2} & T''_{3,3} \\ T_{4,1} & T_{4,2} & T''_{4,3} \\ T_{5,1} & T_{5,2} & T''_{5,3} \\ T_{6,1} & T_{6,2} & T''_{6,3} \end{bmatrix} \begin{pmatrix} v_z^s(0) \\ v_x^s(0) \\ T \end{pmatrix}$$



Applications

Influence of the thickness of the sample



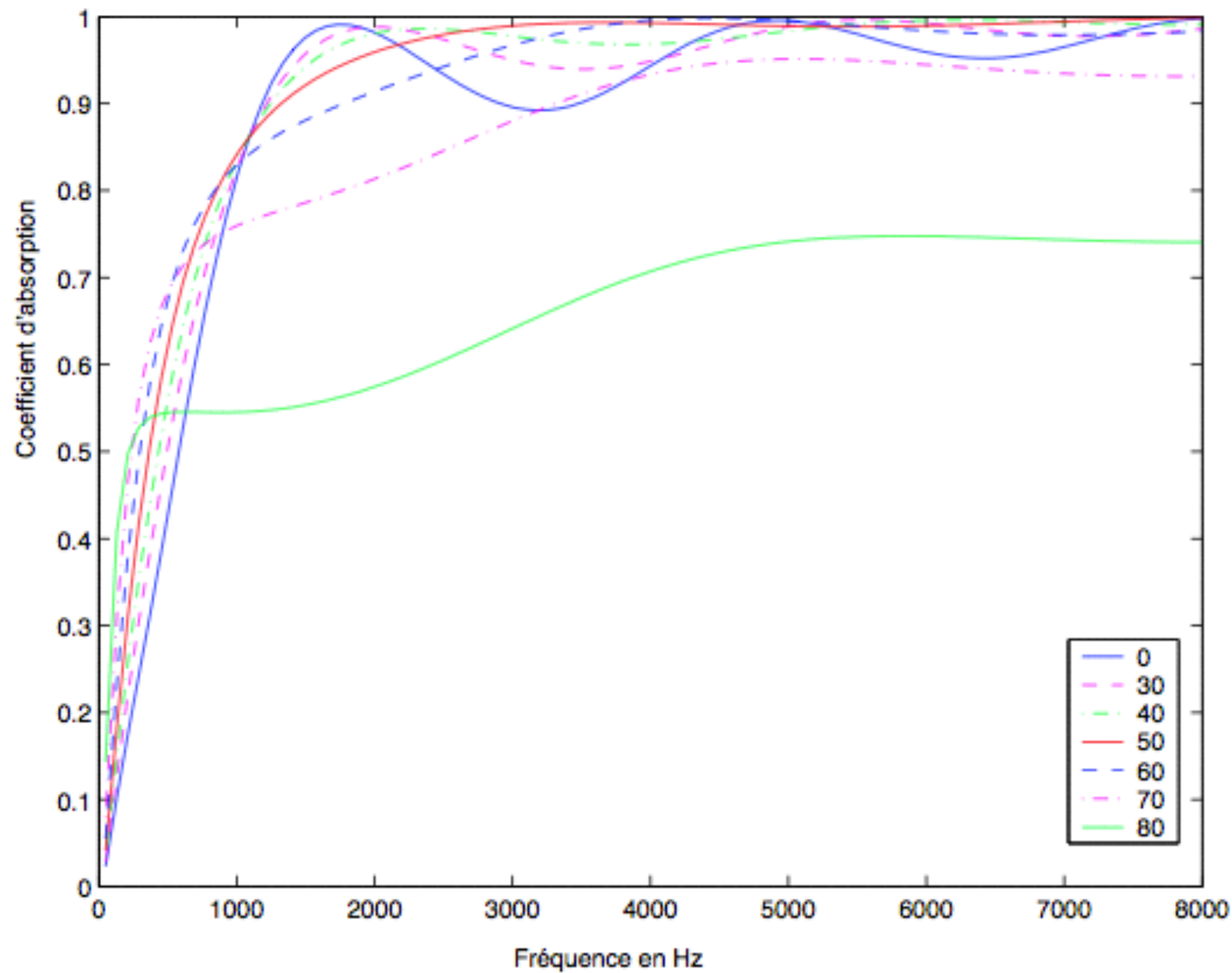
$$\theta = 0 \text{ rad}$$

$$\sigma = 10000 N.m^{-4}.s$$

$$\phi = 0.98$$

$$\alpha_{\infty} = 1$$

Influence of the incident angle



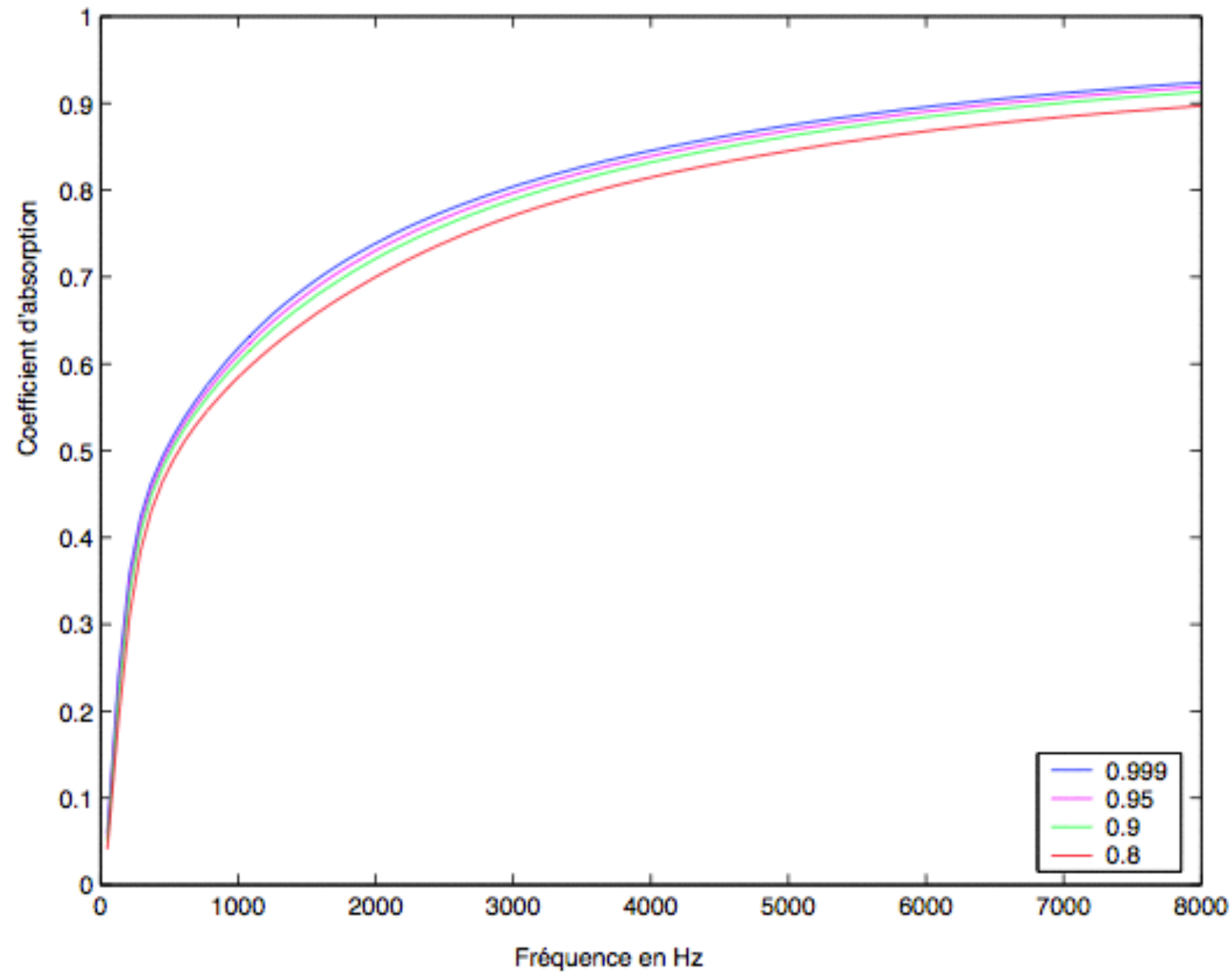
$$l = 50mm$$

$$\sigma = 10000 N.m^{-4}.s$$

$$\phi = 0.98$$

$$\alpha_{\infty} = 1$$

Influence of porosity



$$l = 50mm$$

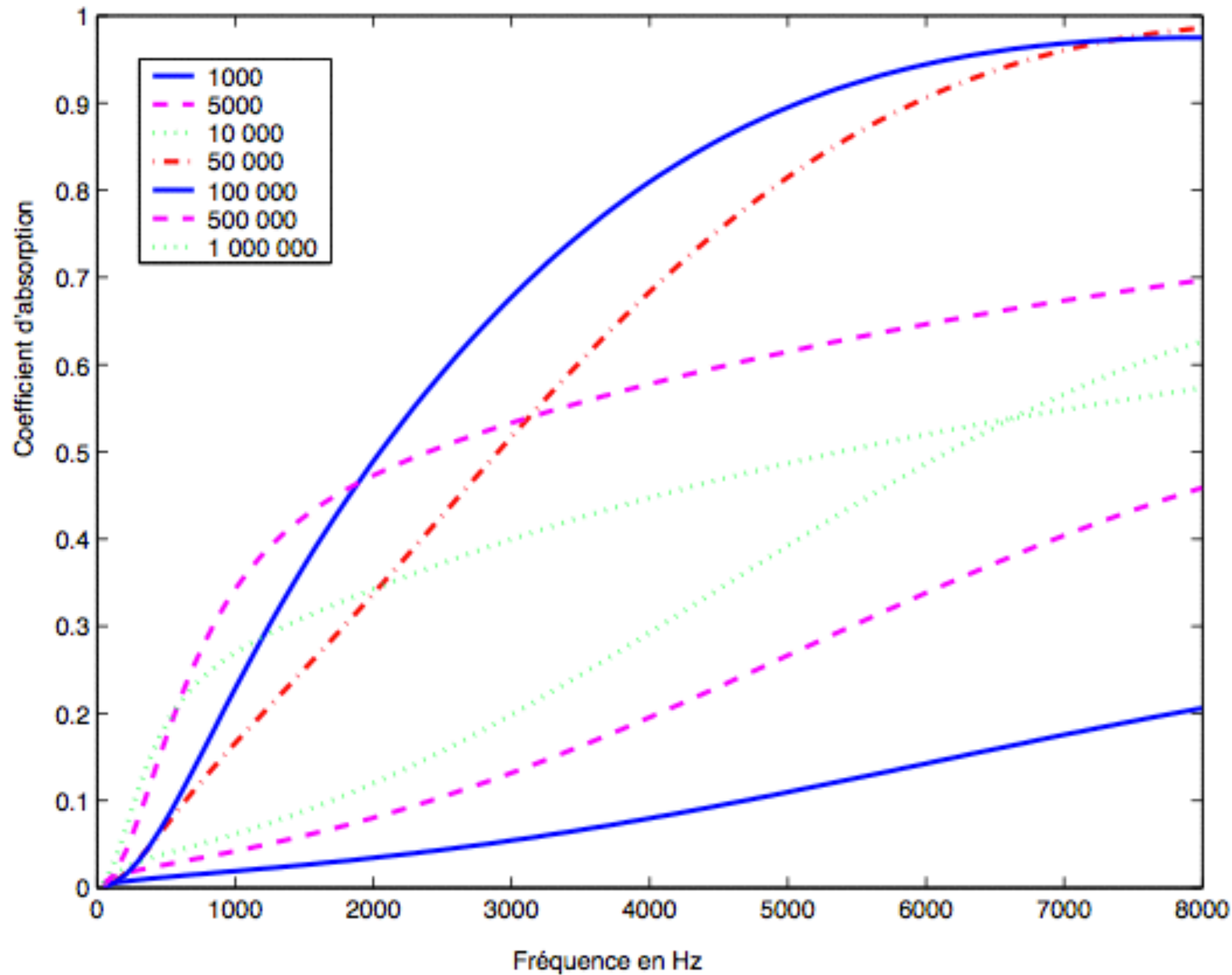
$$\theta = 0 \text{ rad}$$

$$\sigma = 10000 N.m^{-4}.s$$

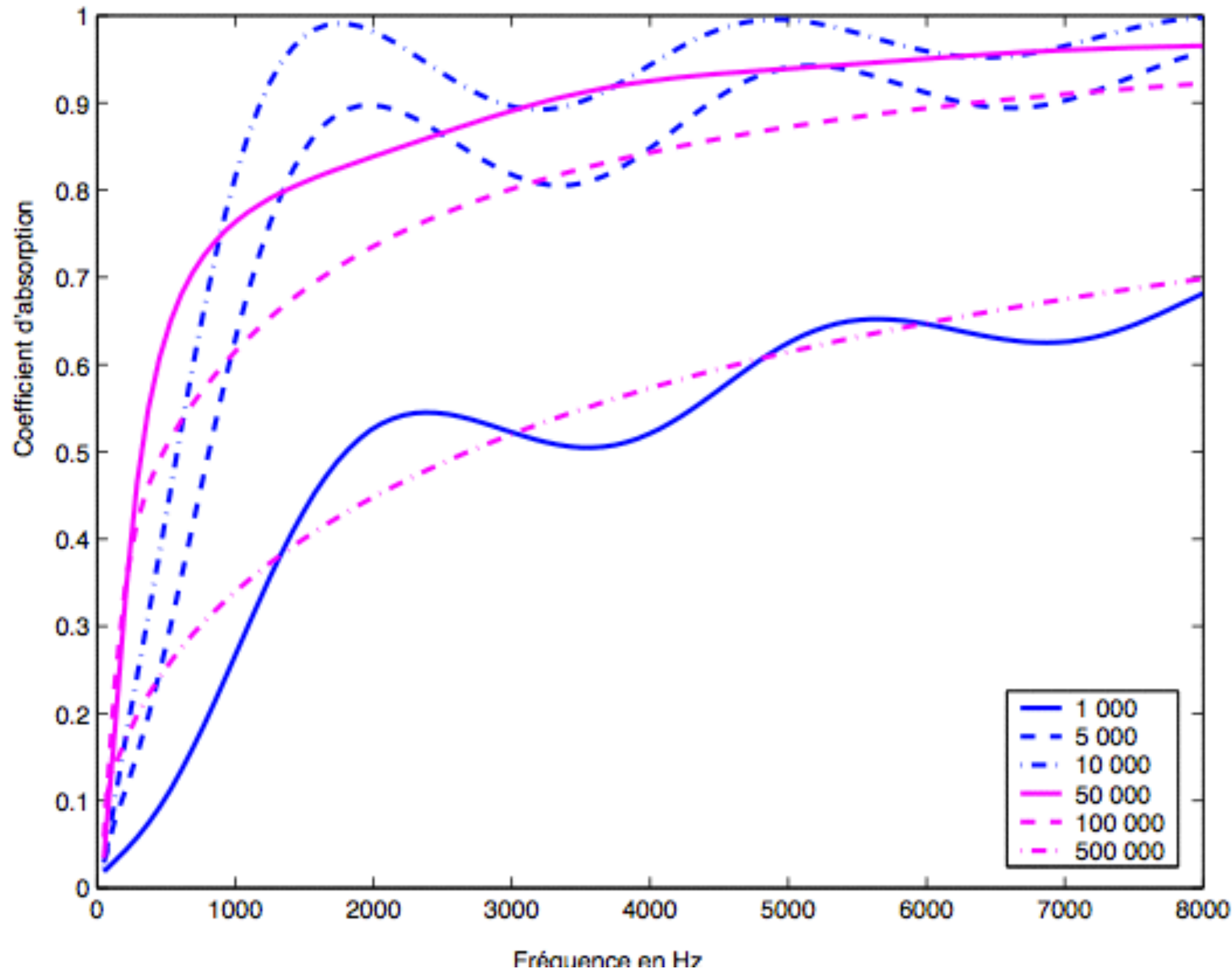
$$\phi = 0.98$$

$$\alpha_{\infty} = 1$$

Influence of resitivity



$$l = 10mm$$
$$\theta = 0 \text{ rad}$$
$$\phi = 0.98$$
$$\alpha_{\infty} = 1$$



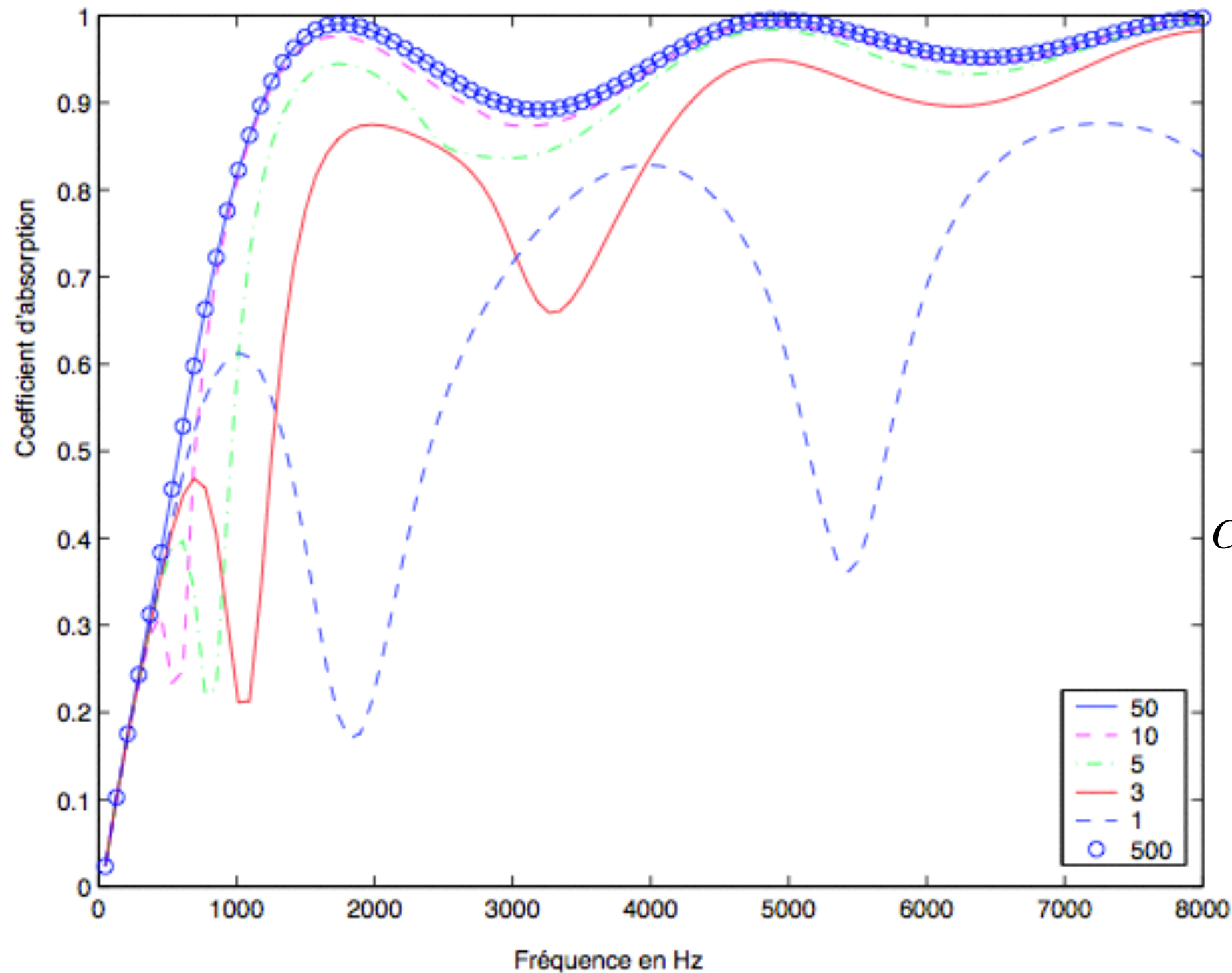
$$l = 50mm$$

$$\theta = 0 \text{ rad}$$

$$\phi = 0.98$$

$$\alpha_{\infty} = 1$$

Influence of the density of the frame



$$l = 50mm$$

$$\sigma = 10000Nm^{-4}s$$

$$\theta = 0 \text{ rad}$$

$$\phi = 0.98$$

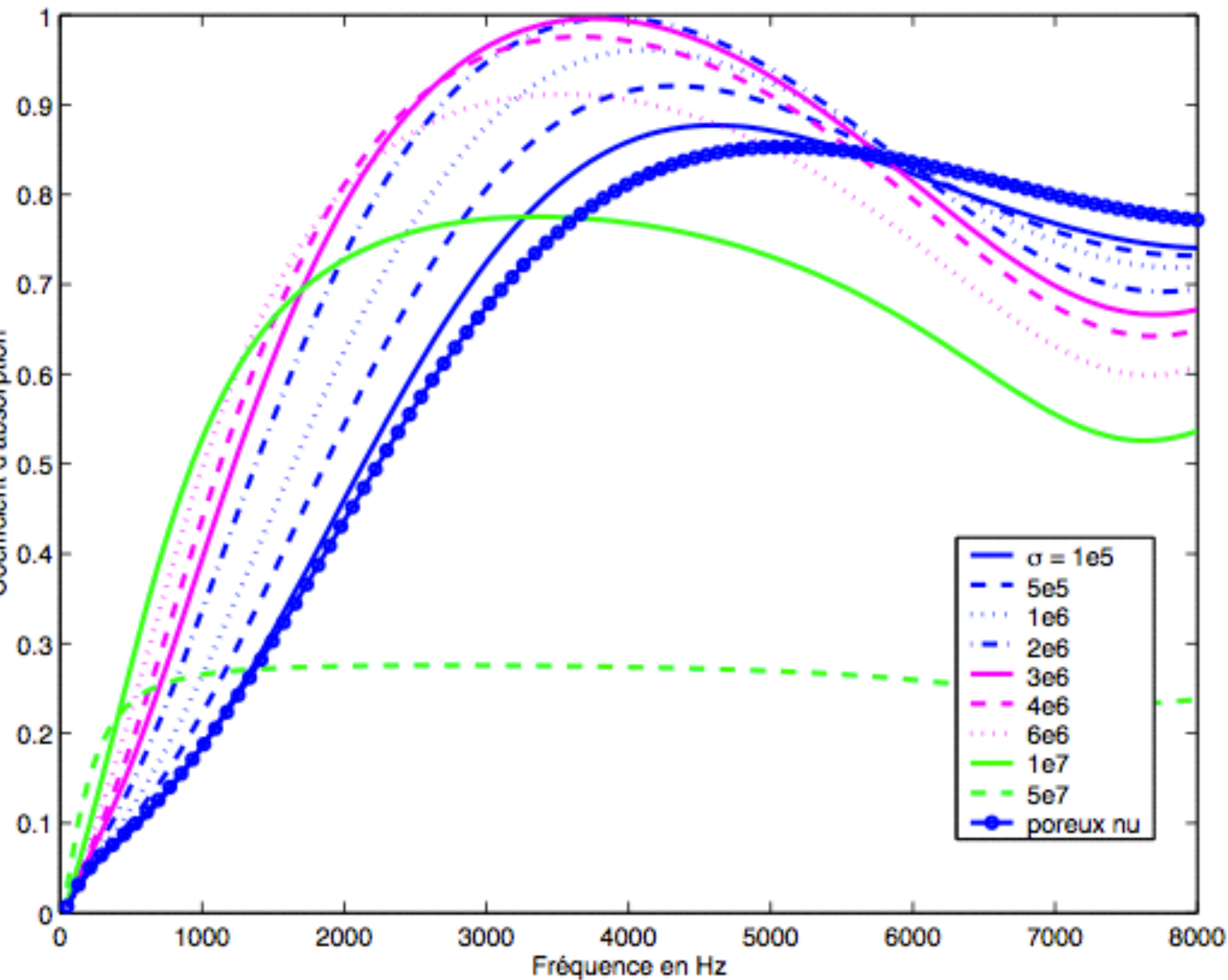
$$\alpha_{\infty} = 1$$

$$E = 10^5 Pa$$

$$\eta = 0.01$$

$$\nu = 0.3$$

Influence of a resistive screen



$$l_s = 0.1mm$$

$$l = 20mm$$

$$\sigma = 10000Nm^{-4}s$$

$$\theta = 0 \text{ rad}$$

$$\phi = 0.98$$

$$\alpha_\infty = 1$$

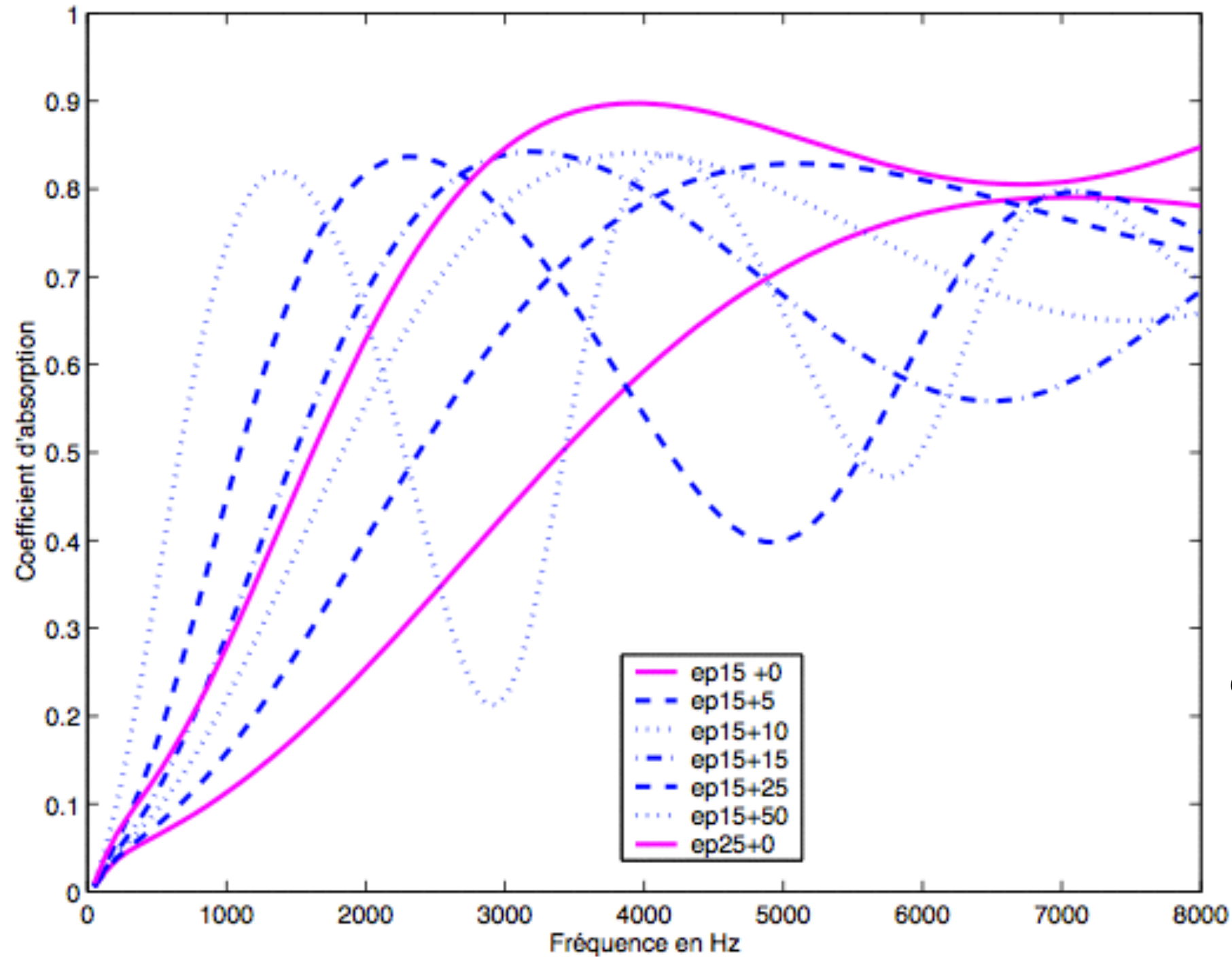
$$E = 2.10^4 Pa$$

$$\eta = 0.05$$

$$\nu = 0$$

$$\rho_1 = 30kgm^{-3}$$

Influence of an air cavity



$$l = 15mm$$

$$\sigma = 10000Nm^{-4}s$$

$$\theta = 0 \text{ rad}$$

$$\phi = 0.98$$

$$\alpha_{\infty} = 1$$